

# *Modeling and Analysis of Dynamic Systems*

## **Vibration of Single-Degree-of-Freedom (SDOF) Systems**

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# Chapter Overview and Learning Objectives

- This Chapter presents the essential concepts in the modeling, analysis, and physics of single-degree-of-freedom (SDOF) vibratory dynamic systems. These form the foundation for more advanced studies.
- Build the foundation in a systematic manner:
  - Modeling (how to derive the governing equations)
  - Understand the basic concepts of natural frequency, resonance, phase shift for linear systems
  - Develop techniques for analytical and numerical solutions for both transient and steady-state responses

# Outline of Topics

1. Modeling and fundamental concepts
2. Free response due to initial disturbances; measurement of damping; MATLAB simulations
3. Forced response to harmonic and periodic excitations; transfer functions; measurement of damping; MATLAB simulations
4. Forced response to non-periodic excitations (impulse, ...); solutions by transform methods
5. Applications (accelerometers, vibration isolation techniques, ...)

# 1. Modeling and Fundamental Concepts

- **Physics:** Understand the concepts of natural frequency (固有频率), damping ratio (阻尼比), and energy dissipation in free vibration
- **Foundation:** Understand the concept of equilibrium in vibratory systems
- **Analysis and Simulations:** Apply Newton's Law and energy method to obtain the governing equation of motion of SDOF systems

# Examples of Vibration

- Simple objects:

Vibration of violin strings, ...

- Structures and Mechanical Systems:

Bridge oscillations due to traffic, earthquake, etc.

Building oscillations due to wind loads

Machinery vibrations due to unbalanced loads

Vibration of diving boards

.....

- How to model and understand these vibration phenomena?



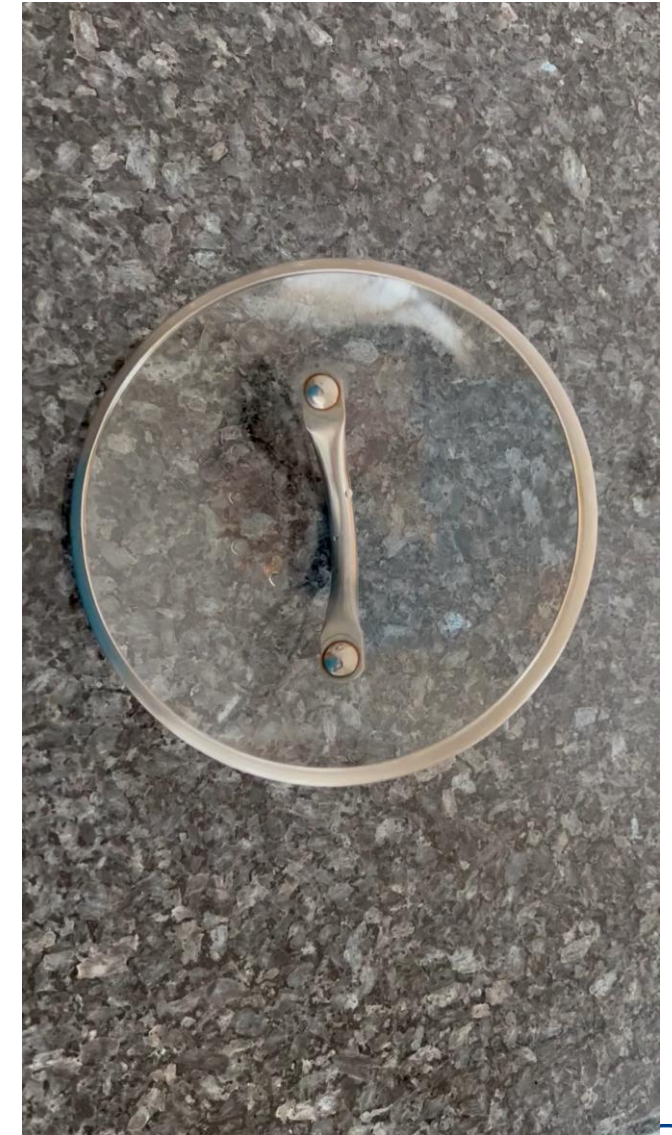
# A Fundamental Thought on Dynamics and Natural Frequency

A key concept that we will study in this class:

Consider the cover of a cooking pot. This is an example of a free vibration (due to an initial disturbance). The pot cover is initially at rest or equilibrium and a small “push” on one edge (along the handle direction) is applied. The cover then oscillates at its natural frequency.

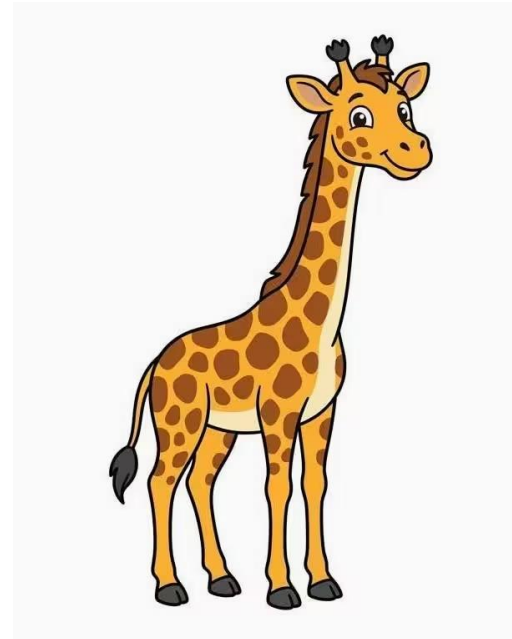
Note: You may think of this like a swinging pendulum.

- What causes the oscillation? What (physics) is going on?
  - An unbalanced force or moment on the system
  - Exchange of kinetic and potential energies of the mass
  - Is there a natural frequency?

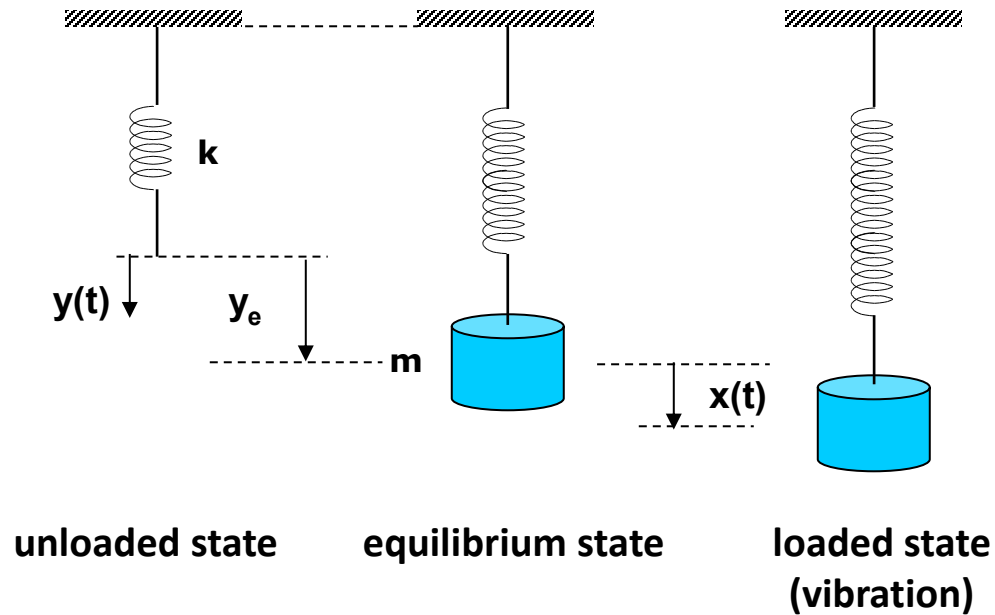


# Vibrations of String-like Systems

- What does the natural frequency of a string depend on?
- The motion of a giraffe neck



# A Prototypical Vibratory System: Use of coordinate systems



$$m\ddot{y} = mg - F_s, \text{ where } F_s = ky$$

$$\Rightarrow m\ddot{y} = mg - k(y_e + x)$$

$$mg = ky_e, \quad y = y_e + x, \quad \ddot{y} = \ddot{x}$$

$$\Rightarrow m\ddot{x} + kx = 0 \text{ or } m_{eq}\ddot{x} + k_{eq}x = 0$$

**This is the basic form of governing equation for the free vibration of SDOF undamped systems**

- Using  $y(t)$  notation: coordinate measured with respect to the unstretched position of the spring (a fixed reference)
- Using  $x(t)$  notation: coordinate measured with respect to the spring equilibrium position
- Formulation (using either  $y(t)$  or  $x(t)$ ) does not alter the fundamental response characteristics, i.e., gravity does not affect the formulation.
- Free vibration: no external force, motion introduced by initial disturbances

## Derive governing equation for undamped system by energy method:

22-13. The body of arbitrary shape has a mass  $m$ , mass center at  $G$  and a radius of gyration about  $G$  of  $k_G$ . If it is displaced a slight amount  $\theta$  from its equilibrium position and released, determine the natural period of vibration.

Derive governing eqn of motion by energy principle:

$$T = \frac{1}{2} I_O \dot{\theta}^2, \quad V = -mgd \cos\theta$$

$$E = T + V; \quad \frac{dE}{dt} = 0$$

$$\Rightarrow \frac{d}{dt} \left( \frac{1}{2} I_O \dot{\theta}^2 - mgd \cos\theta \right) = 0$$

$$\Rightarrow I_O \dot{\theta} \ddot{\theta} + mgd \sin\theta \dot{\theta} = 0 \Rightarrow I_O \ddot{\theta} + mgd \sin\theta = 0$$

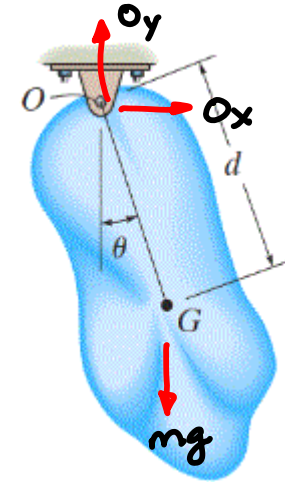
Apply the parallel axis theorem,  $I_O = \underbrace{mk_G^2}_{I_G} + md^2$

The governing equation can be written in the standard form for small angle of oscillation

$$\ddot{\theta} + \underbrace{\frac{mgd}{I_O}}_{\omega_o^2} \theta = 0$$

Thus, the natural frequency  $\omega_o = \sqrt{\frac{mgd}{I_O}}$ . The natural period of vibration is

$$T_o = \frac{2\pi}{\omega_o} = 2\pi \sqrt{\frac{I_O}{mgd}} = 2\pi \sqrt{\frac{mk_G^2 + md^2}{mgd}} = 2\pi \sqrt{\frac{k_G^2 + d^2}{gd}}$$



Natural frequency is a function of material properties.

The uniform rod of mass  $m$  is supported by a pin at  $A$  and a spring at  $B$ . If  $B$  is given a small sideward displacement and released, determine the natural period of vibration.

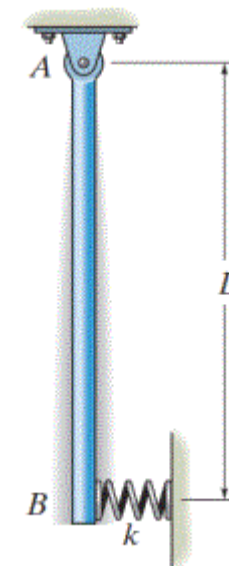
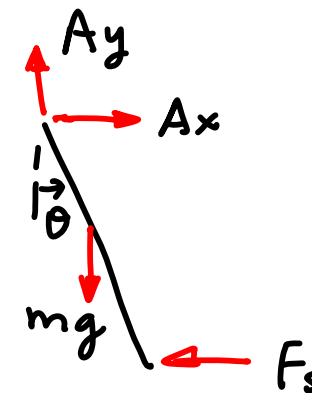
Assume the spring is unstretched when the rod is in the vertical position shown. Consider an angular displacement  $\theta$  and the moment eqn about the fixed point  $A$ ,

$$\boxed{\sum M_A = I_A \alpha} ; \quad I_A = \frac{ml^2}{3}$$

$$\Rightarrow -mg \frac{l}{2} \sin\theta - F_s l \cos\theta = I_A \ddot{\theta}$$

$$\text{where } F_s = k l \sin\theta$$

$$\Rightarrow -\frac{mgl}{2} \sin\theta - kl^2 \sin\theta \cos\theta = \frac{1}{3} ml^2 \ddot{\theta}$$



For small angles (so that the spring stays horizontal),  $\sin\theta \approx \theta$ ,  $\cos\theta \approx 1$

Thus, the governing eqn is:

$$-\frac{mgl}{2}\theta - kl^2\theta = \frac{1}{3}ml^2\ddot{\theta}$$

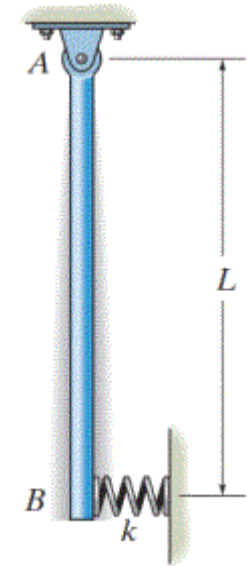
$$\Rightarrow \frac{1}{3}ml^2\ddot{\theta} + \left(\frac{mgl}{2} + kl^2\right)\theta = 0 \quad (1)$$

Write Eqn (1) in standard form

$$\ddot{\theta} + \left(\frac{3mg + 6kl}{2ml}\right)\theta = 0 \Rightarrow \omega_o = \sqrt{\frac{3mg + 6kl}{2ml}}$$

$$\text{Natural period of oscillation, } T_o = \frac{2\pi}{\omega_o} = 2\pi \sqrt{\frac{2ml}{3mg + 6kl}}$$

Note that, in general, natural frequency (period) increases (decreases) with the stiffness. Hence, stiffer violin strings produce higher pitch frequencies.



Derive the governing eqn of motion by the energy method (for conservative systems).

Kinetic energy,  $T = \frac{1}{2} I_A \dot{\omega}^2$  ;  $\omega = \dot{\theta}$

Potential energy,  $V = -mg\frac{l}{2}\cos\theta + \frac{1}{2}k(l\sin\theta)^2$   
 $\uparrow$  datum at level of A

$E = T + V = \text{Constant}$ . Applying  $\frac{dE}{dt} = 0$  leads to:

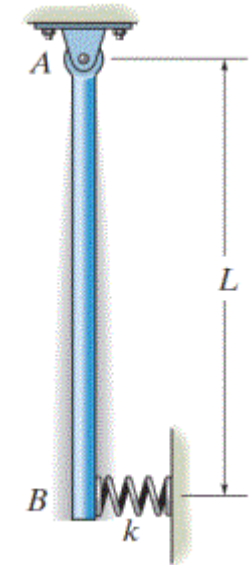
$$\frac{dE}{dt} = I_A \ddot{\theta} \dot{\theta} + \frac{mg l}{2} \sin\theta \dot{\theta} + k l^2 \sin\theta \cos\theta \dot{\theta} = 0$$

For small angles,  $\sin\theta \approx \theta$ ,  $\cos\theta \approx 1$

$$\Rightarrow \left[ I_A \ddot{\theta} + \left( \frac{mg l}{2} + k l^2 \right) \theta \right] \dot{\theta} = 0$$

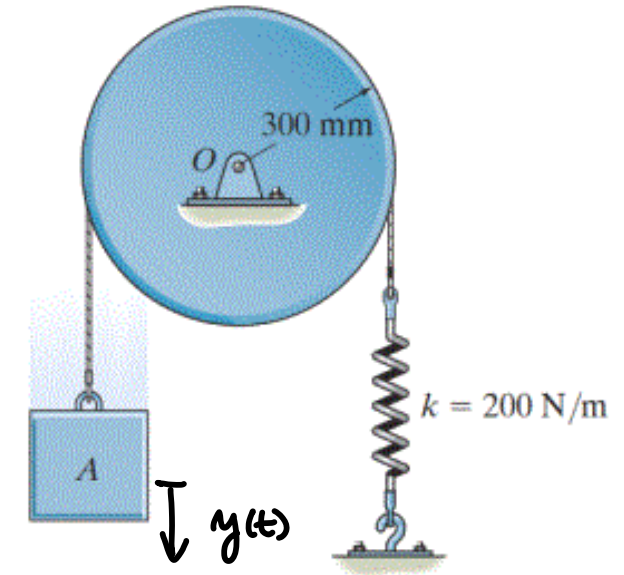
Set this equal to zero  
to obtain eqn of motion

$$\Rightarrow \frac{1}{3} m l^2 \ddot{\theta} + \left( \frac{mg l}{2} + k l^2 \right) \theta = 0$$



The energy method, whenever applicable, is easier to apply.

The 20kg disk, is pinned at its mass center  $O$  and supports the 4kg block  $A$ . If the belt which passes over the disk is not allowed to slip at its contacting surface, determine the natural period of vibration of the system.



### Energy Method :

Let  $y(t)$  be the displacement of mass  $A$  measured from the position when the spring is unstretched.

$$T = \frac{1}{2} I_O \dot{\theta}^2 + \frac{1}{2} m_A \dot{y}^2 \quad \text{no-slip} \Rightarrow y = R\theta, \dot{y} = R\dot{\theta}$$

$$V = -m_A g y + \frac{1}{2} k y^2 = -m_A g R\theta + \frac{1}{2} k (R\theta)^2$$

$$\begin{aligned} \frac{dE}{dt} = 0 &\Rightarrow I_O \ddot{\theta} \dot{\theta} + m_A R^2 \ddot{\theta} \dot{\theta} - m_A g R \dot{\theta} + k R^2 \theta \dot{\theta} = 0 \\ &\Rightarrow (I_O + m_A R^2) \ddot{\theta} + k R^2 \theta = m_A g R \end{aligned} \quad (1)$$

Physics explained: There is an equilibrium position for which the spring is stretched due to the weight. Set the time derivative terms in (1) to zero (eqm means  $\ddot{\theta} = 0$ )

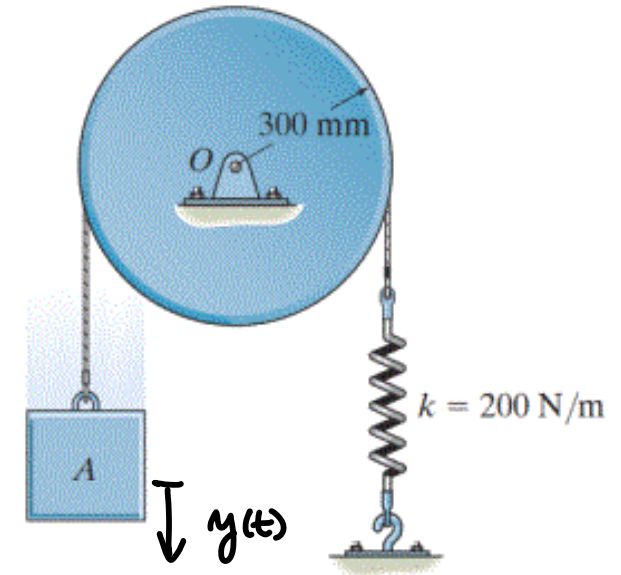
$$\Rightarrow kR^2\theta_{eq} = m_A g R \quad (2)$$

$$\Rightarrow \theta_{eq} = m_A g / k R \quad (\text{Spring force balanced by } m_A g)$$

Define  $\beta = \theta - \theta_{eq}$  (angle of oscillation measured from the equilibrium position)

Eqn (1) then becomes

$$(I_0 + m_A R^2) \ddot{\beta} + kR^2 \beta = 0 \Rightarrow \omega_0 = \left[ \frac{kR^2}{\frac{1}{2}m_D R^2 + m_A R^2} \right]^{1/2} = 3.78 \text{ rad/s}$$



### Newton's Law :

Establish the FBD for the two bodies (the disk and the mass)

$F_t$  = tension force in rope

$$\text{mass: } m_A g - F_t = m_A \ddot{y} = m_A R \ddot{\theta} \quad (3)$$

$$\begin{aligned} \text{disk: } \sum M_o &= I_o \ddot{\theta} \\ \Rightarrow F_t R - F_s R &= I_o \ddot{\theta} \end{aligned} \quad (4)$$

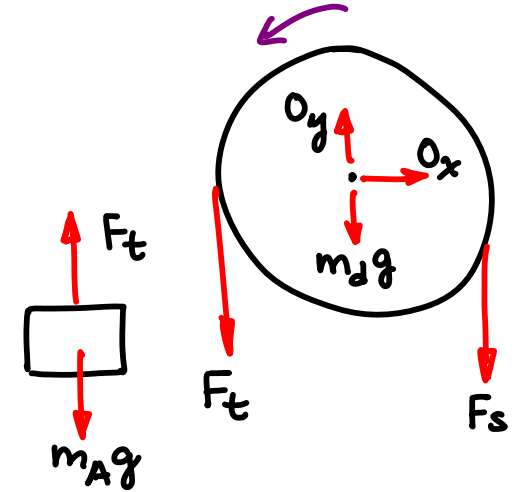
Eliminate  $F_t$  from Eqns (3) and (4) leads to :

$$m_A g R - F_s R = (I_o + m_A R^2) \ddot{\theta} \quad , \quad F_s = k y = k R \theta$$

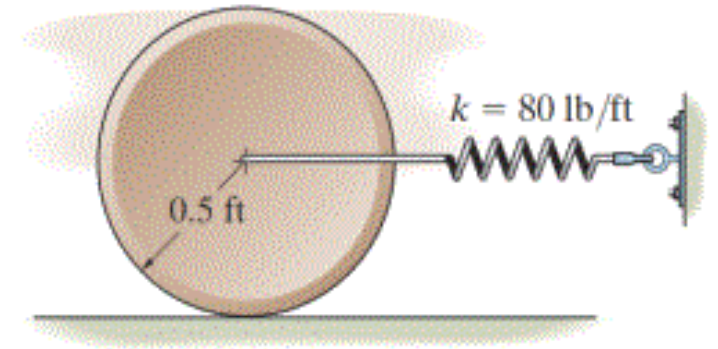
$$\Rightarrow (I_o + m_A R^2) \ddot{\theta} + k R^2 \theta = m_A g R \quad \leftarrow \text{Eqn. (1) derived earlier}$$

Physics of vibration:

$$\omega_o = \left[ \frac{k R^2}{\frac{1}{2} m_d R^2 + m_A R^2} \right]^{1/2} \Rightarrow \text{natural frequency increases with stiffness and decreases with mass/inertia}$$



The disk has a weight of 30 lb and rolls without slipping on the horizontal surface as it oscillates about its equilibrium position. If the disk is displaced, by initially rolling it counterclockwise 0.2 rad and rotating it clockwise 1 rad/s, determine the equation which describes its oscillatory motion and the natural period when it is released.



Energy method:

$$\begin{aligned} T &= \frac{1}{2} I_G \dot{\theta}^2 + \frac{1}{2} m v_G^2 \\ &= \frac{1}{2} I_G \dot{\theta}^2 + \frac{1}{2} m \underbrace{(R\dot{\theta})^2}_{\text{no-slip}} \end{aligned}$$

$$V = \frac{1}{2} k \Delta x^2 = \frac{1}{2} k (R\theta)^2$$

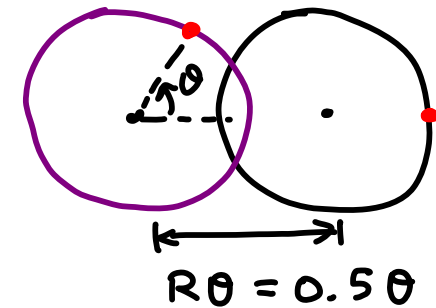
$$E = T + V = \text{constant} \quad (\text{for conservative systems})$$

$$\frac{dE}{dt} = 0 \Rightarrow (I_G \ddot{\theta} + mR^2 \ddot{\theta} + kR^2 \theta) \dot{\theta} = 0$$

$$\Rightarrow (I_G + mR^2) \ddot{\theta} + kR^2 \theta = 0 \quad \text{is the eqn of motion}$$

$$\text{The natural frequency } \omega_o = \sqrt{\frac{kR^2}{I_G + mR^2}}, \quad I_G = \frac{1}{2} mR^2$$

$$\Rightarrow \omega_o = \sqrt{\frac{2k}{3m}} = 7.566 \text{ rad/s}; \quad T_o = 2\pi/\omega_o = 0.83 \text{ sec}$$

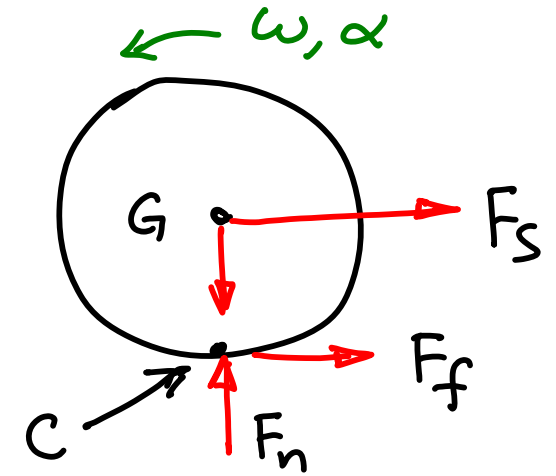


Next, we will apply the basic Newton's Law to obtain the governing eqn of motion. We will see that the energy method is generally much simpler.

Newton's Law:

Draw FBD as shown.

Note that, by using the positive convention of counterclockwise rotation,  $F_f$  must be in the direction shown.



$$\begin{aligned}\sum M_C &= I_G \alpha + (\vec{r}_{G/C} \times m \vec{a}_G) \cdot \hat{k} \\ \Rightarrow -F_s R &= I_G \alpha + (R \hat{j} \times m R \alpha (-\hat{i})) \cdot \hat{k} \quad , \quad \alpha = \ddot{\theta} \\ \Rightarrow -(kR\theta)R &= I_G \alpha + mR^2 \alpha \quad \xrightarrow{\text{assume no-slip}} \vec{a}_G = -R\alpha \hat{i} \\ \Rightarrow (I_G + mR^2) \ddot{\theta} + kR^2 \theta &= 0\end{aligned}$$

The above is the same eqn as above.

Note that if we consider the moment eqn at G, the procedure is more involved.

$$\sum M_G = I_G \alpha$$

$$\Rightarrow F_f R = I_G \alpha \quad \leftarrow \text{need an eqn to eliminate } F_f \text{ from the formulation}$$

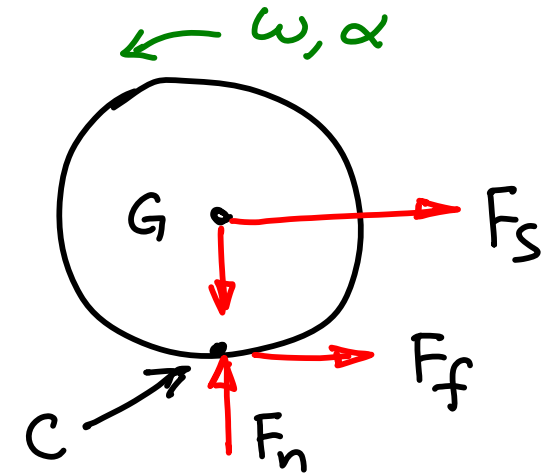
$$\sum F_x = m a_{Gx} \Rightarrow F_s + F_f = m(-R\alpha)$$

$$\Rightarrow F_f = -mR\alpha - F_s$$

$$\text{Thus, } (-mR\alpha - F_s)R = I_G \alpha$$

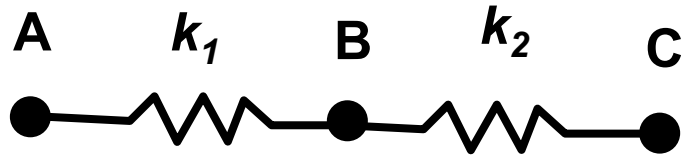
$$\Rightarrow -F_s R = I_G \alpha + mR^2 \alpha \quad \leftarrow \text{same eqn as above}$$

Note: if one needs to prove no-slip, then  $F_f$  needs to be solved to test the hypothesis  $|F_f| \leq \mu_s F_n$ .



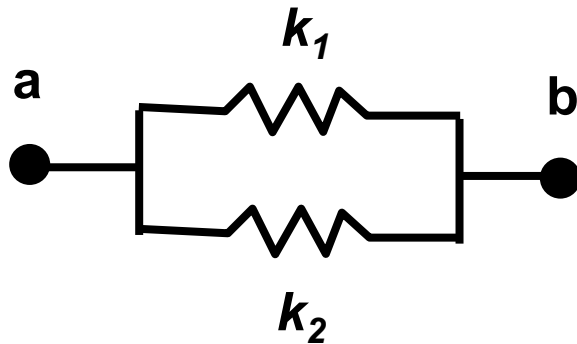
# Systems with Combined Springs

- Equivalent spring constants



series:  $k_{AC} = \frac{1}{\frac{1}{k_1} + \frac{1}{k_2}}$

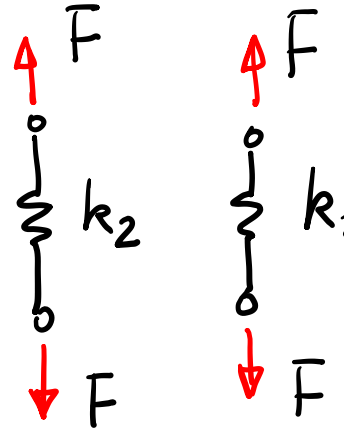
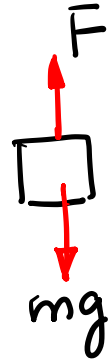
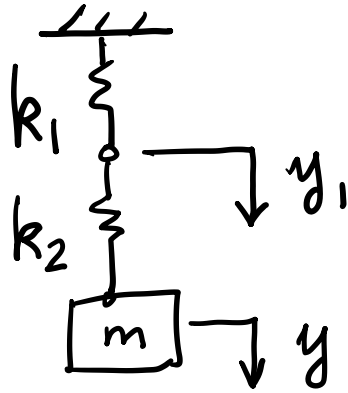
$$\frac{1}{k_{AC}} = \frac{1}{k_1} + \frac{1}{k_2}$$



parallel:  $k_{ab} = k_1 + k_2$

- This is identical to the combination of capacitors in electrical circuits

## Springs in series :



$y, y_1$  are absolute coordinates

understand why all these forces are  $F$ .  
(Newton's 3rd law)

$$\begin{aligned} mg - F &= m\ddot{y} \\ F &= k_1 y_1 \Rightarrow y_1 = F/k_1 \\ F &= k_2 (y - y_1) \end{aligned} \quad \left. \vphantom{\begin{aligned} mg - F &= m\ddot{y} \\ F &= k_1 y_1 \Rightarrow y_1 = F/k_1 \\ F &= k_2 (y - y_1) \end{aligned}} \right\} F = k_2 \left( y - \frac{F}{k_1} \right)$$

$$mg - F = m\ddot{y}$$

$$\left. \begin{array}{l} F = k_1 y_1 \Rightarrow y_1 = F/k_1 \\ F = k_2 (y - y_1) \end{array} \right\} F = k_2 \left( y - \frac{F}{k_1} \right)$$

Simplify to get:  $F = \frac{k_1 k_2}{k_1 + k_2} y$

$$\Rightarrow mg - \frac{k_1 k_2}{k_1 + k_2} y = m\ddot{y} \Rightarrow m\ddot{y} + \underbrace{\left( \frac{k_1 k_2}{k_1 + k_2} \right)}_{k_{eq}} y = mg$$

Define  $x = y - y_e$ , where  $k_e y_e = mg$

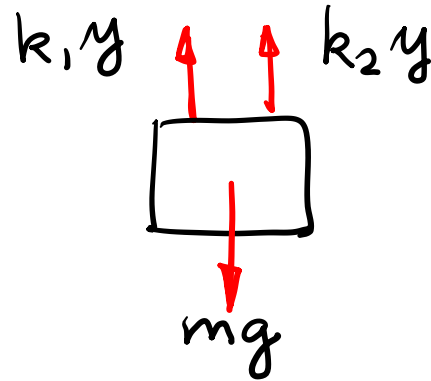
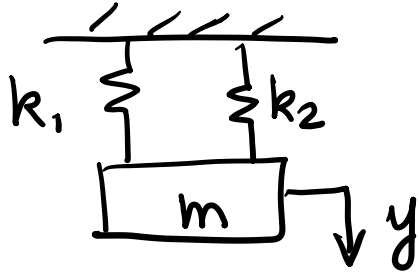
$$\Rightarrow m\ddot{x} + k_{eq} x = 0 \quad (\text{standard eqn of motion})$$

$$k_{eq} = \frac{k_1 + k_2}{k_1 + k_2}$$

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

Note that  $k_{eq}$  is independent of gravity

Spring in parallel :



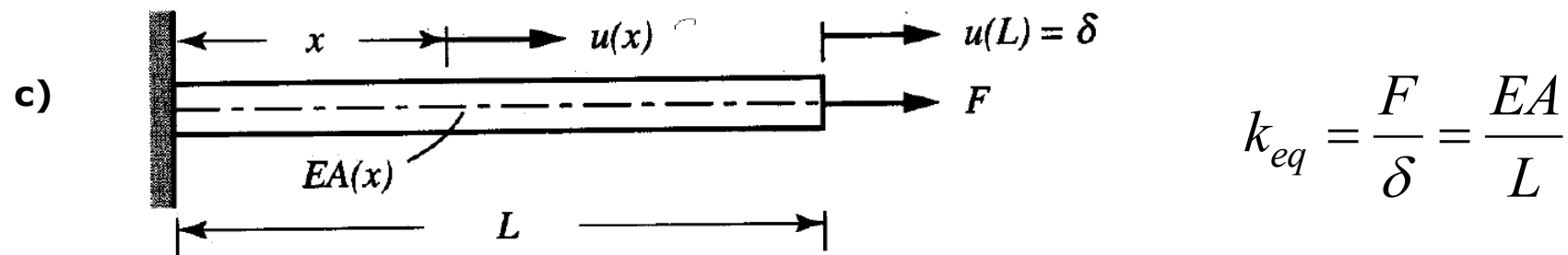
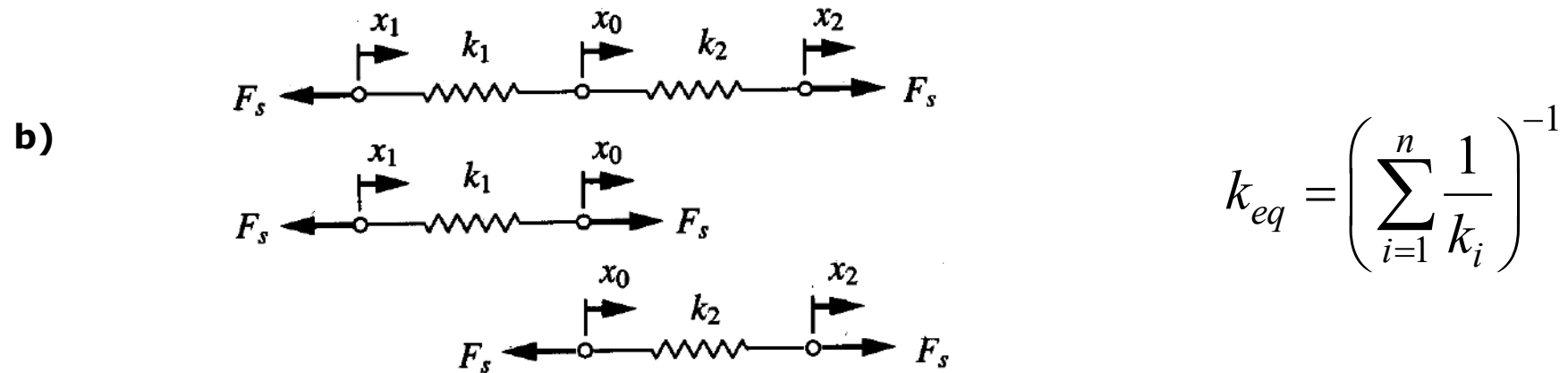
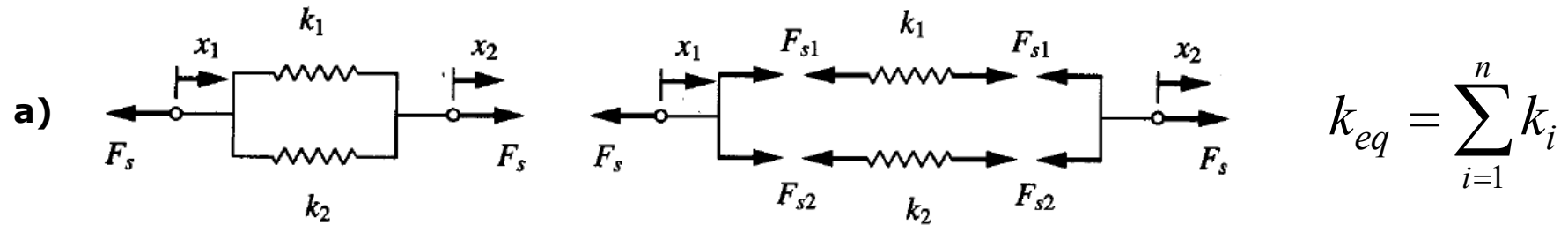
$$m\ddot{y} = mg - (k_1 + k_2)y$$

$$x = y - y_e, \quad 0 = mg - (k_1 + k_2)y_e$$

$$\Rightarrow m\ddot{x} + \underbrace{(k_1 + k_2)}_{k_{eq}} x = 0$$

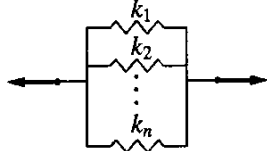
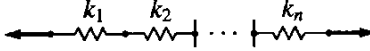
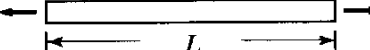
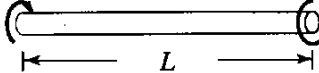
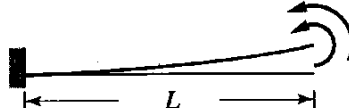
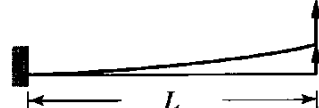
$$k_{eq} = k_1 + k_2$$

# Equivalent discrete elements



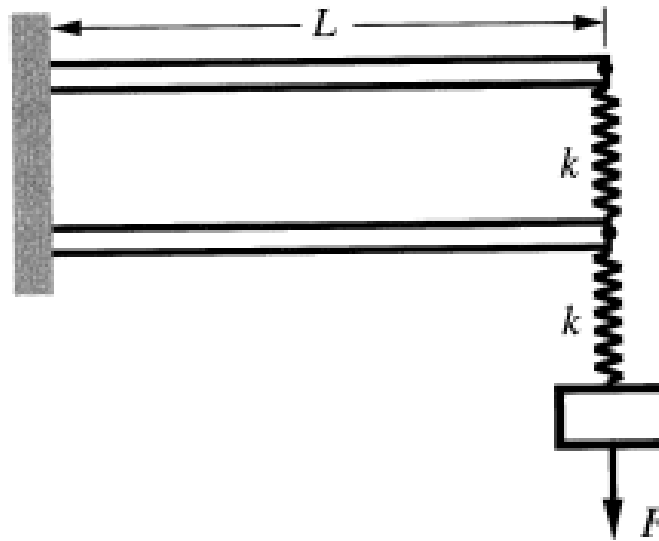
# Table of equivalent stiffness models

- Meirovitch *Fundamentals of Vibration*, Table 1.2 on equivalent spring constants

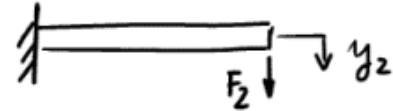
| Table 1.2 - Equivalent Spring Constants                                                                                                                                                              |                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Component Sketch and Description                                                                                                                                                                     | $k_{eq}$                         |
|  <p>Springs in parallel</p>                                                                                       | $\sum_{i=1}^n k_i$               |
|  <p>Springs in series</p>                                                                                         | $\frac{1}{\sum_{i=1}^n (1/k_i)}$ |
|  <p>Torsional spring</p>                                                                                          | $\frac{EI}{L}$                   |
|  <p>Rod in axial deformation</p>                                                                                  | $\frac{EA}{L}$                   |
|  <p>Shaft in torsion</p>                                                                                          | $\frac{GJ}{L}$                   |
|  <p>Helical spring<br/> <math>d</math> = diameter of coil cross section<br/> <math>n</math> = number of coils</p> | $\frac{Gd^4}{64nR^3}$            |
|  <p>Cantilever beam with a moment at the tip</p>                                                                 | $\frac{EI}{L}$                   |
|  <p>Cantilever beam with a force at the tip</p>                                                                 | $\frac{3EI}{L^3}$                |

# Equivalent Stiffness Example 1:

[Ginsberg 1.3] Identical cantilevered elastic beams having flexural rigidity  $EI$  are combined with two identical springs  $k$  to support the block. Determine the static displacement of the block due to a static force  $F$  and deduce the equivalent stiffness of the system.



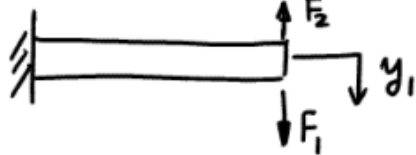
### Ginberg 1.3



Consider the statics problem.  
 $F_2$  = force in upper spring

$$F_2 = k_{\text{beam}} y_2 = \frac{3EI}{L^3} y_2$$

also,  $F_2 = k \Delta_2$   
 $\Delta_2$  = net deflection in the upper spring



$$y_1 = y_2 + \Delta_2$$

$$= \frac{F_2}{3EI/L^3} + \frac{F_2}{k}$$

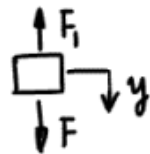
$$\Rightarrow F_2 = \frac{k(3EI/L^3)}{k + 3EI/L^3} y_1$$

lower beam:  $F_1 - F_2 = \frac{3EI}{L^3} y_1$

$$\Rightarrow F_1 = \left[ \frac{3EI}{L^3} + \frac{k(3EI/L^3)}{k + 3EI/L^3} \right] y_1$$

$$= \frac{3EI}{L^3} \left[ \frac{2k + 3EI/L^3}{k + 3EI/L^3} \right] y_1$$

neglect weight  
 or consider  
 deflection w.r.t.  
 eqm position



Lower beam analysis:  $F_1 = k \Delta_1$ , where  $F_1 = F$

$$y = y_1 + \Delta_1$$

Simplify above relations,

$$\begin{aligned} y &= \frac{F}{\frac{3EI}{L^3} \left[ \frac{2k + 3EI/L^3}{k + 3EI/L^3} \right]} + \frac{F}{k} \\ &= \underbrace{\left[ \frac{k + 3EI/L^3}{\frac{3EI}{L^3} [2k + 3EI/L^3]} + \frac{1}{k} \right]}_{\frac{1}{k_{eq}}} F \quad (\text{springs in series}) \end{aligned}$$

Simplify (denote  $\frac{3EI}{L^3}$  as  $k_b$ , for simplicity)

$$\begin{aligned} \frac{1}{k_{eq}} &= \frac{k + k_b}{k_b (2k + k_b)} + \frac{1}{k} \\ &= \frac{k^2 + k k_b + 2k k_b + k_b^2}{k k_b (2k + k_b)} \end{aligned}$$

$$\Rightarrow k_{eq} = \frac{k k_b (2k + k_b)}{k^2 + 3k k_b + k_b^2}$$

$$\Rightarrow \boxed{k_{eq} = \frac{3kEI/L^3 (2k + 3EI/L^3)}{k^2 + 9kEI/L^3 + (3EI/L^3)^2}}$$

# Concept of natural frequency for free undamped response

To solve  $m\ddot{x} + kx = 0$ , seek a solution of the form:

$$x(t) = A \cos(\omega_n t - \phi)$$

where  $A$  is the amplitude and  $\phi$  is the phase angle.

Differentiating with respect to time twice gives:

$$\dot{x}(t) = -\omega_n A \sin(\omega_n t - \phi)$$

$$\ddot{x}(t) = -\omega_n^2 A \cos(\omega_n t - \phi) = -\omega_n^2 x(t)$$

Substituting back into the equation of motion gives:

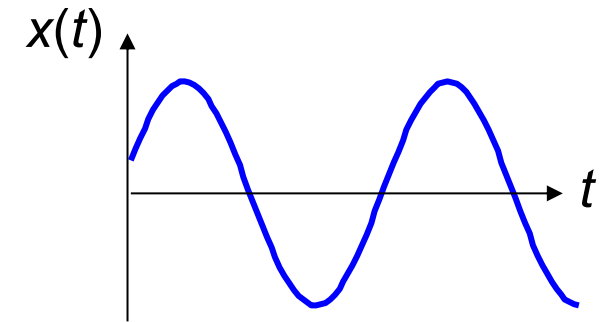
$$(-m\omega_n^2 + k) A \cos(\omega_n t - \phi) = 0$$

Since  $A \neq 0$ , simplifying the above gives the frequency equation:

$$-m\omega_n^2 + k = 0 \text{ or } \omega_n = \sqrt{k/m}$$

$$\text{Period: } T_n = 2\pi/\omega_n$$

**Natural frequency (the frequency at which the system naturally vibrates). This is a characteristic of the system.**



simple harmonic motions

# Concept of natural frequency for free undamped response

Alternatively, seek solution of the form more formal approach):

$$x(t) = Ae^{\lambda t}$$

where  $\lambda$  is called the eigenvalue or characteristic value. Thus,

$$\ddot{x} = \lambda^2 Ae^{\lambda t}$$

Note that:

$$\cos \omega_n t = \frac{e^{i\omega_n t} + e^{-i\omega_n t}}{2}$$

$$e^{i\omega_n t} = \cos \omega_n t + i \sin \omega_n t$$

An advantage of this approach is that the derivative terms are easier.

$$m\ddot{x} + kx = 0, \quad \Rightarrow (m\lambda^2 + k)Ae^{\lambda t} = 0$$

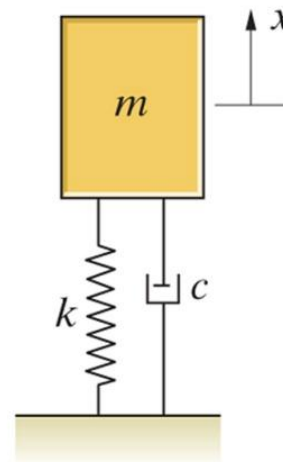
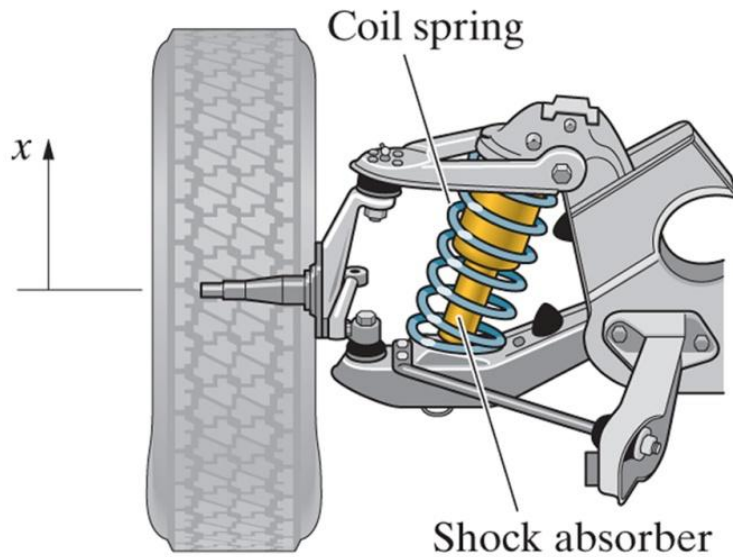
For non-trivial solutions, the following equation (the so-called frequency equation) must be satisfied:

$$m\lambda^2 + k = 0 \Rightarrow \lambda_{1,2} = \pm\sqrt{k/m} i = \pm\omega_n i$$

The fundamental solutions for  $e^{\pm i\omega_n t}$  are sine and cosine functions with frequency of  $\omega_n$ . Thus, the two approaches are both valid. However, this latter approach usually leads to simpler algebra.

# Modeling of SDOF Systems with Damping

- Consider the suspension modeled as a SDOF system
  - tire: inertia or mass
  - coil spring: stiffness or structural flexibility
  - shock absorber: viscous damping or energy dissipation



Governing equation of motion:

$$m\ddot{x} + c\dot{x} + kx = 0$$

or,

$$m_{eq}\ddot{x} + c_{eq}\dot{x} + k_{eq}x = 0$$

Bedford and Fowler, *Engineering Dynamics*, 5th edition

## 2. Free Response of SDOF Systems

- Derive the time response solutions for both undamped and damped SDOF dynamic systems

Governing equation – solution describes evolution of motion

$$m_{eq}\ddot{x} + c_{eq}\dot{x} + k_{eq}x = 0$$

Initial conditions – how the motion starts

$$x(t = 0) = x_0, \quad \dot{x}(0) = v_0$$

- Understand the fundamental physics (phase shift, energy exchange, ...)
- Numerical solutions

# General solution for the undamped ( $\zeta = 0$ ) system:

Seek a solution of the form:

$$x(t) = A \cos(\omega_n t - \phi), A = \text{amplitude}, \phi = \text{phase angle}$$

where  $A$  is the amplitude and  $\phi$  is the phase angle.

Differentiating with respect to time twice gives:

$$\dot{x}(t) = -\omega_n A \sin(\omega_n t - \phi)$$

$$\ddot{x}(t) = -\omega_n^2 A \cos(\omega_n t - \phi) = -\omega_n^2 x(t)$$

Substituting back into the equation of motion gives:

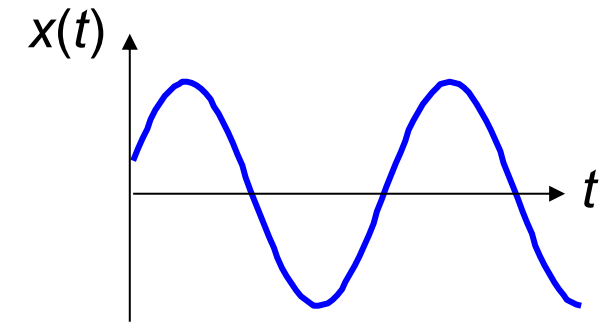
$$(-m\omega_n^2 + k) A \cos(\omega_n t - \phi) = 0$$

Simplifying the above gives the frequency equation:

$$-m\omega_n^2 + k = 0 \text{ or } \omega_n = \sqrt{k/m} \leftarrow$$

**Natural frequency (the frequency at which the system naturally vibrates). This is a characteristic of the system.**

$$\text{Period: } T_n = 2\pi/\omega_n$$



simple harmonic motions

# Concept of initial conditions

If a system is vibrating then we must assume that something must have (in the past) transferred energy into to the system and caused it to move. For example the mass could have been:

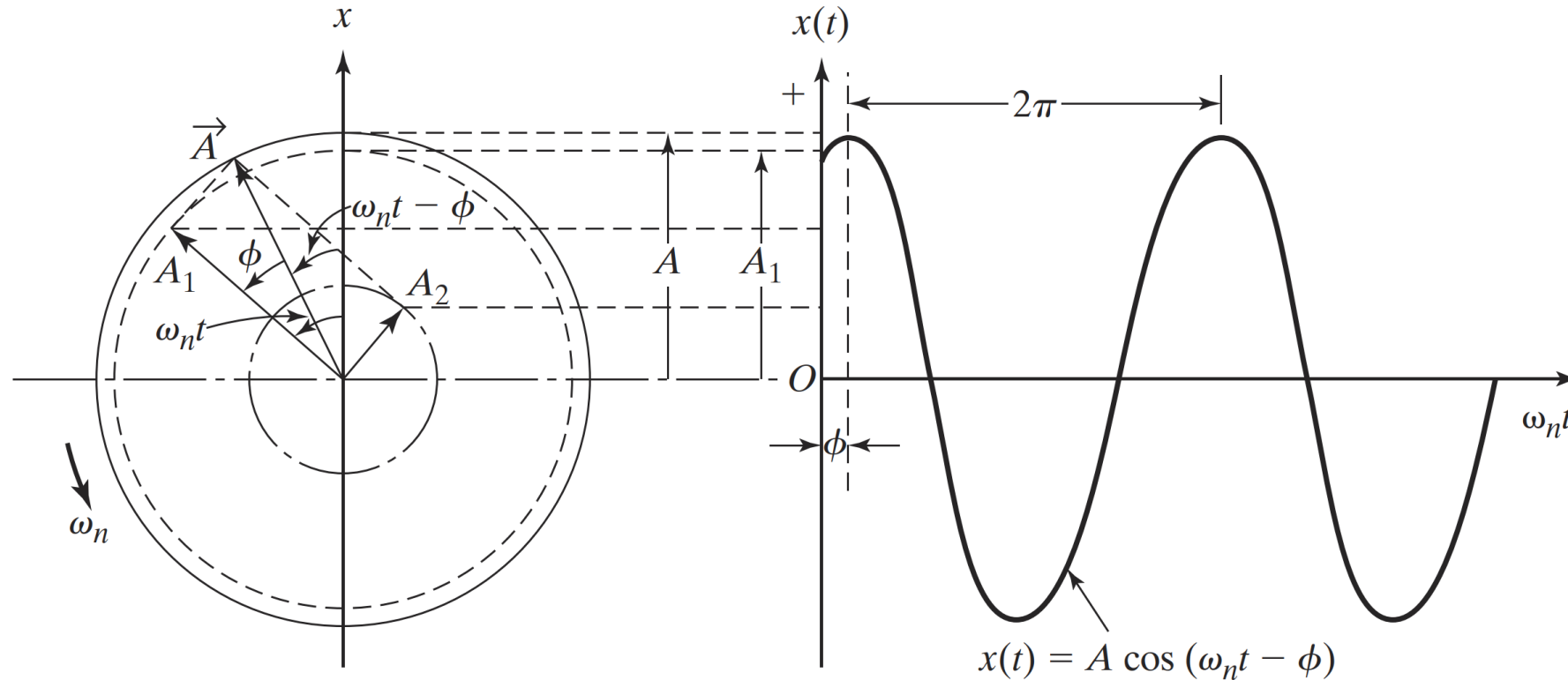
- moved a distance  $x_0$  and then released at  $t = 0$  (i.e., given an initial potential energy) or
- given an initial velocity  $v_0$  (e.g., given an initial kinetic energy due to impact) or
- some combination of the two above cases

Mathematically speaking, initial conditions must be specified in order to have unique solution to the ODE (to solve for  $A$  and  $\phi$ ). From our earlier solution, we can write:

$$x(0) = x_0 = A \cos(\omega_n 0 - \phi) = A \cos \phi$$

$$\dot{x}(0) = v_0 = -\omega_n A \sin(\omega_n 0 - \phi) = \omega_n A \sin \phi$$

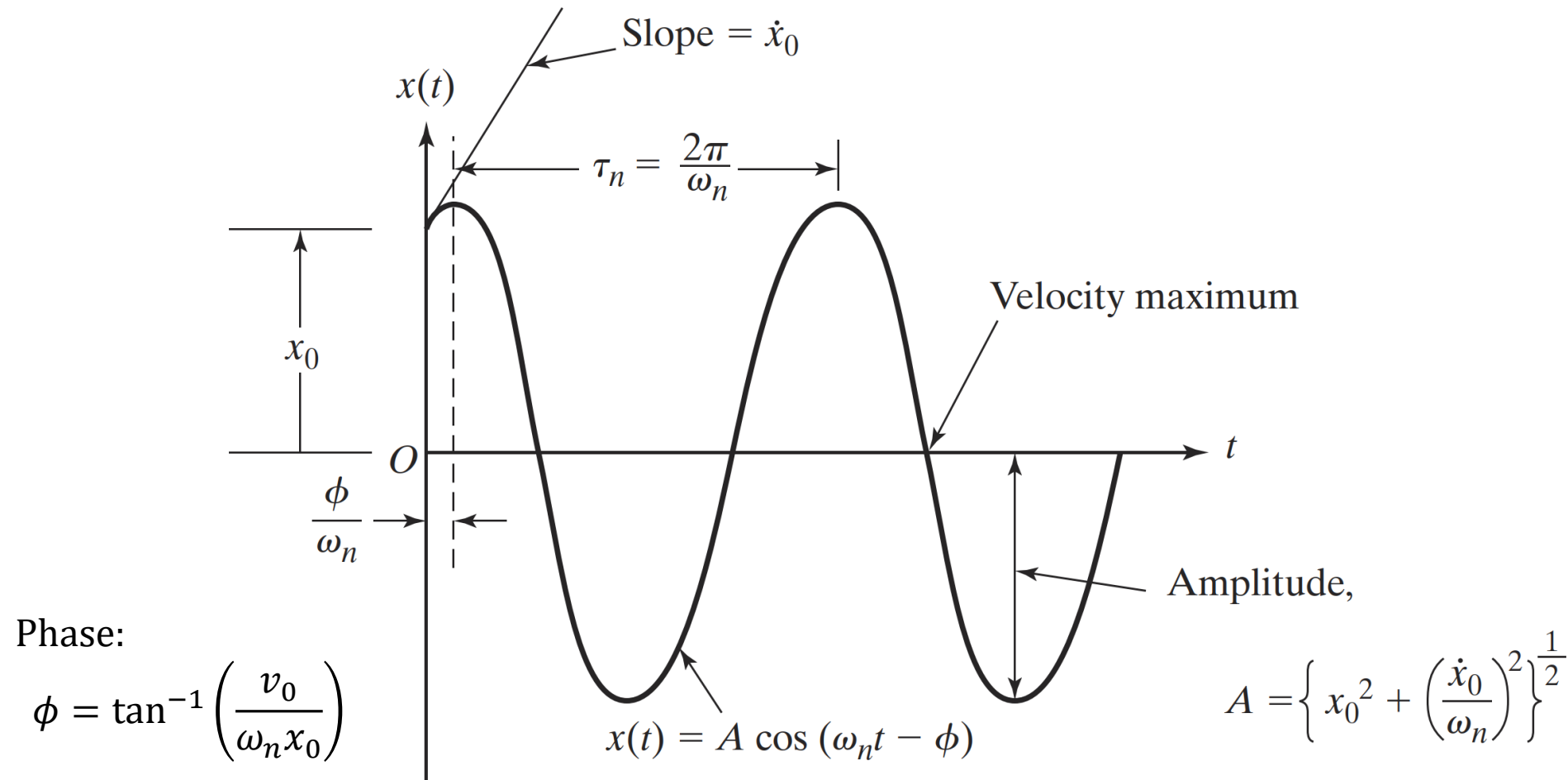
# Characteristics of simple harmonic motions (using the cosine form)



Rao, *Mechanical Vibrations*,  
6th edition, 2018.

$$f_n = \frac{\omega_n \text{ in rad/s}}{2\pi \text{ in rad/cycle}} = \frac{\omega_n}{2\pi} \text{ cycles/s} = \frac{\omega_n}{2\pi} \text{ Hz}$$

# Characteristics of simple harmonic motions (using the cosine form)



## Another representation of the general solution (sine form):

Seek a solution of the form:

$$x(t) = A \sin(\omega_n t + \psi)$$

Mathematically the same solution but with different phase angle. Thus,

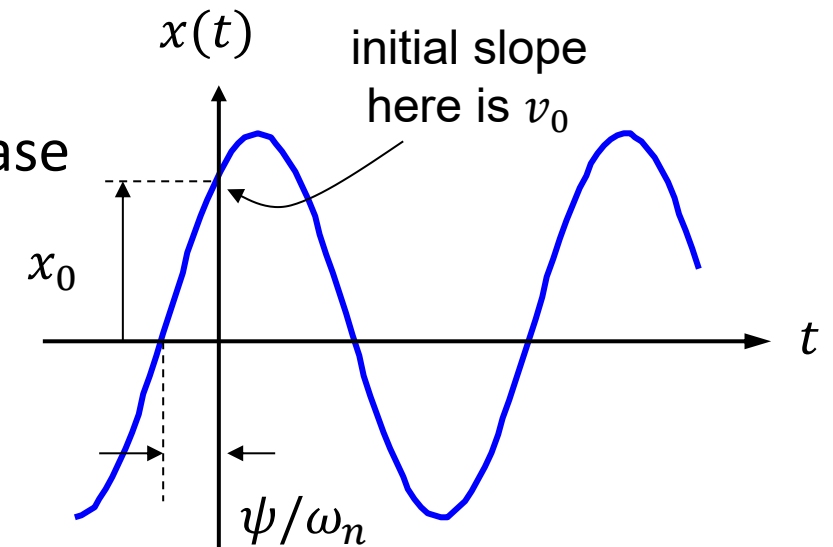
$$x_0 = x(0) = A \sin(\omega_n 0 + \psi) = A \sin \psi$$

$$v_0 = \dot{x}(0) = \omega_n A \cos(\omega_n 0 + \psi) = \omega_n A \cos \psi$$

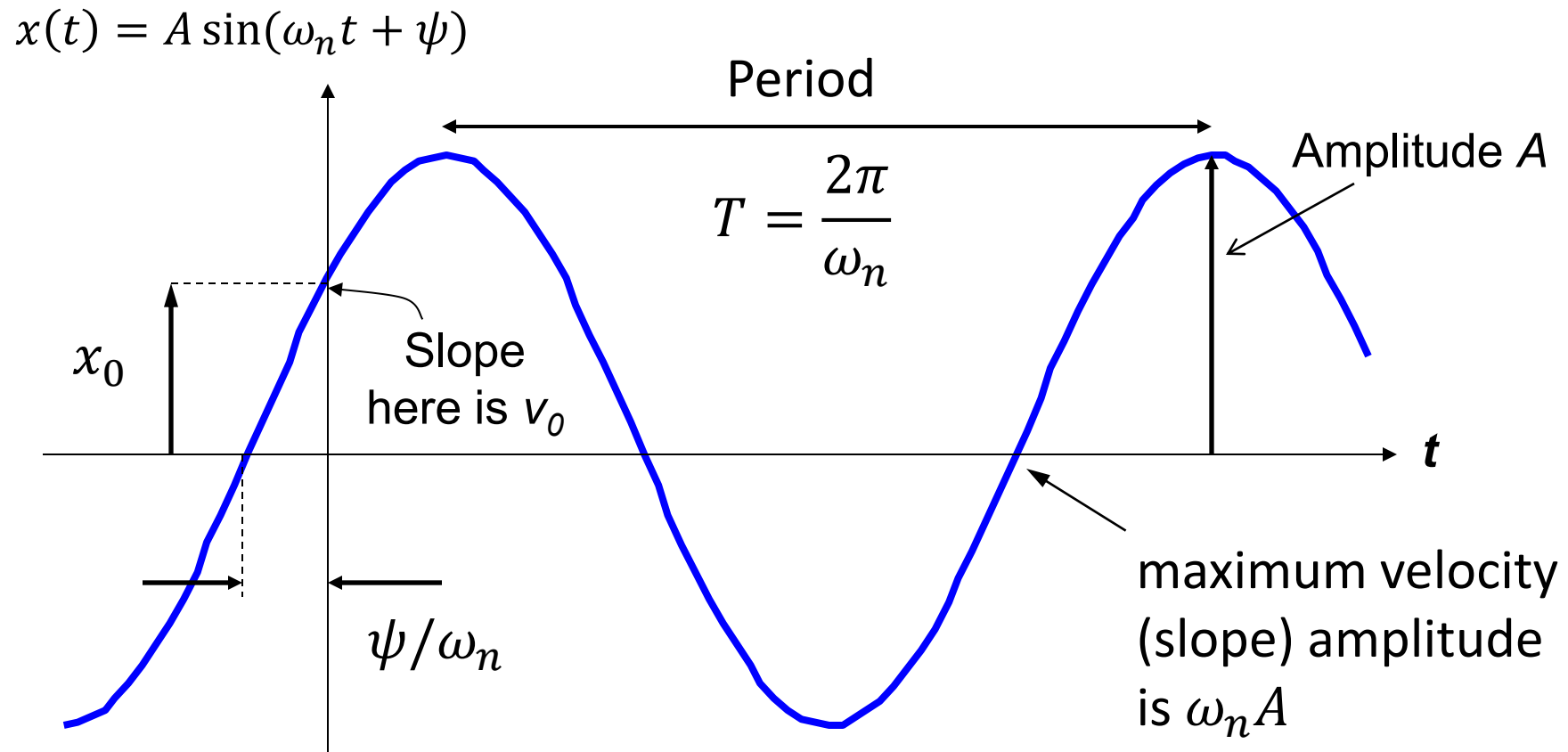
Solving to obtain:

$$A = \frac{1}{\omega_n} \sqrt{\omega_n^2 x_0^2 + v_0^2}, \quad \psi = \tan^{-1} \left( \frac{\omega_n x_0}{v_0} \right)$$

Note that  $(\tan \phi) (\tan \psi) = 1$ , hence  $\phi + \psi = \pi/2$



# Characteristics of simple harmonic motions (using the sine form)



$$f_n = \frac{\omega_n \text{ in rad/s}}{2\pi \text{ in rad/cycle}} = \frac{\omega_n}{2\pi} \text{ cycles/s} = \frac{\omega_n}{2\pi} \text{ Hz}$$

# Characteristics of simple harmonic motions (using the sine form)

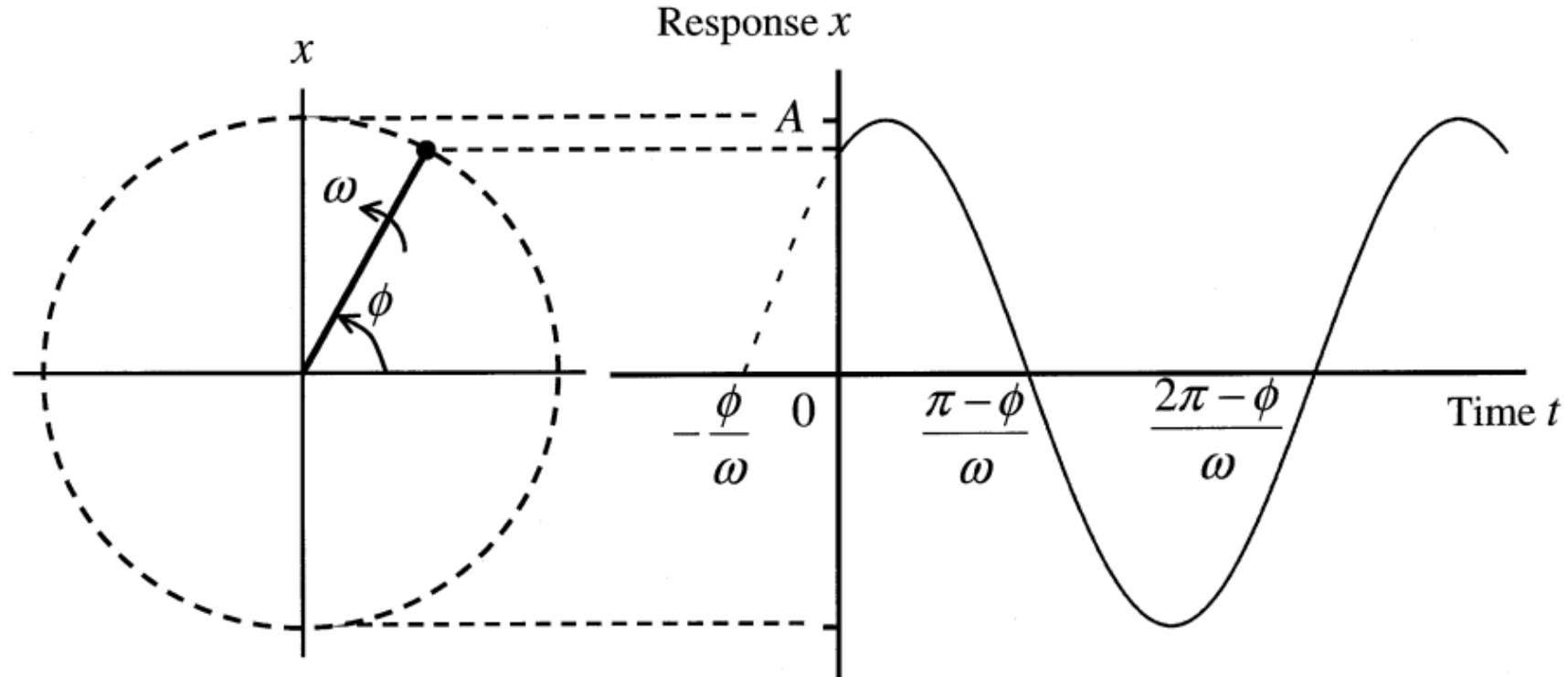
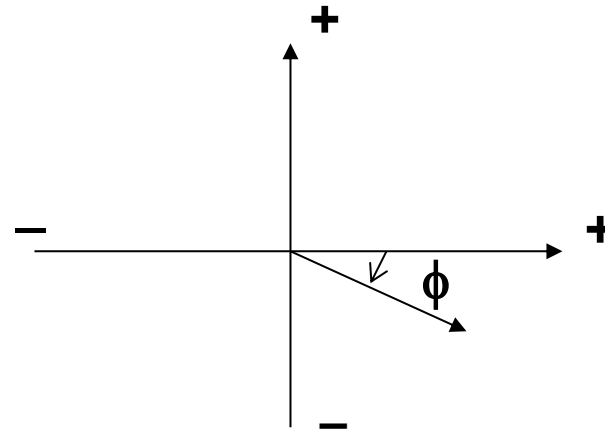
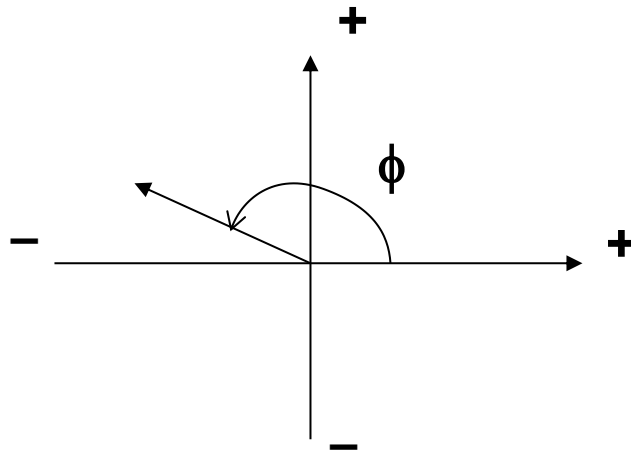


FIGURE 2.6 Free response of an undamped simple oscillator.

$\phi$  should  
be denoted  
as  $\psi$

# A note on arctangents

- Calculating arctangent from a calculator requires attention. First, all machines work in radians.
- The argument  $\text{atan}(-/+)$  is in a different quadrant than  $\text{atan}(+/-)$ , and usual machine calculations will return an arctangent in between  $-\pi/2$  and  $+\pi/2$ , reading only the  $\text{atan}(\cdot)$  for both of the above two cases.



In MATLAB, use the  $\text{atan2}(x,y)$  function to obtain the correct phase.

## Finding the maximum amplitudes for undamped SDOF systems

For undamped systems, the concept of energy conservation, i.e., energy is conserved throughout the motion, can be applied to find the maximum amplitudes. Given initial conditions,  $x(t = 0) = x_0$  and  $\dot{x}(t = 0) = v_0$ , we have

$$E = \frac{1}{2}k_{eq}x_0^2 + \frac{1}{2}m_{eq}v_0^2 = \text{constant}$$

Maximum response amplitude occurs when the velocity is zero (apply the derivative concept) or zero KE. In other words, the PE is maximized when the KE is zero.

$$E = \frac{1}{2}k_{eq}x_0^2 + \frac{1}{2}m_{eq}v_0^2 = \frac{1}{2}k_{eq}A_{max}^2 \Rightarrow A_{max} = \frac{1}{\omega_n} \sqrt{\omega_n^2 x_0^2 + v_0^2}$$

Using the same argument, amplitude of maximum velocity occurs when the KE is maximized or when the PE is zero. Hence  $|v_{max}|$  can be calculated *without taking a derivative*.

$$E = \frac{1}{2}k_{eq}x_0^2 + \frac{1}{2}m_{eq}v_0^2 = \frac{1}{2}m_{eq}v_{max}^2 \Rightarrow v_{max} = \sqrt{\omega_n^2 x_0^2 + v_0^2}$$

These formulas, based on analysis of undamped systems are useful for engineering applications because they are reasonable conservative estimates especially for lightly damped systems.

## In-class exercise:

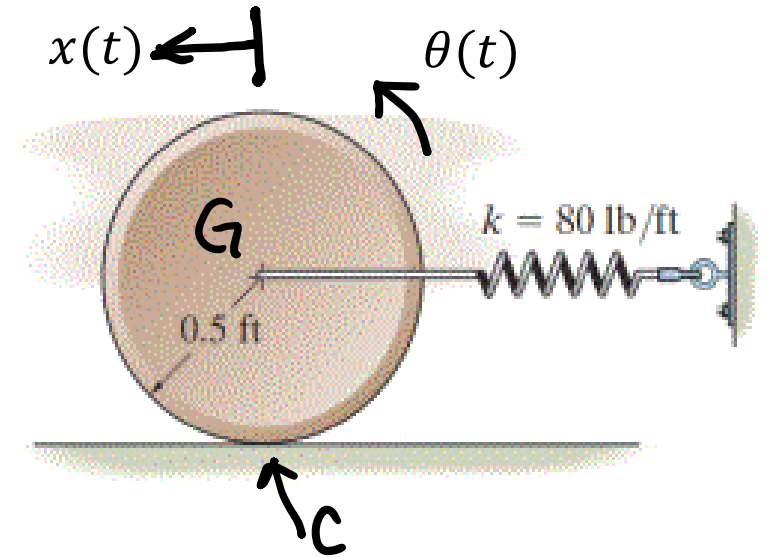
The disk has a weight of 30 lb and rolls without slipping on the horizontal surface as it oscillates about its equilibrium position. If the disk is displaced, by initially rolling it counterclockwise 0.2 rad and rotating it clockwise with 1 rad/s.

**Question:** Find the maximum angular velocity of the disk during the ensuing oscillation.

**Hint:** Apply principle of conservation of energy for this undamped system.

**Notes:**  $m = \frac{30}{32.2}$  lbm or slug (standard English unit)

$$T = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} I_G \dot{\theta}^2, x = r\theta, I_G = \frac{mr^2}{2}$$



## In-class exercise:

The disk has a weight of 30 lb and rolls without slipping on the horizontal surface as it oscillates about its equilibrium position. If the disk is displaced, by initially rolling it counterclockwise 0.2 rad and rotating it clockwise with 1 rad/s.

**Question:** Find the maximum angular velocity of the disk during the ensuing oscillation.

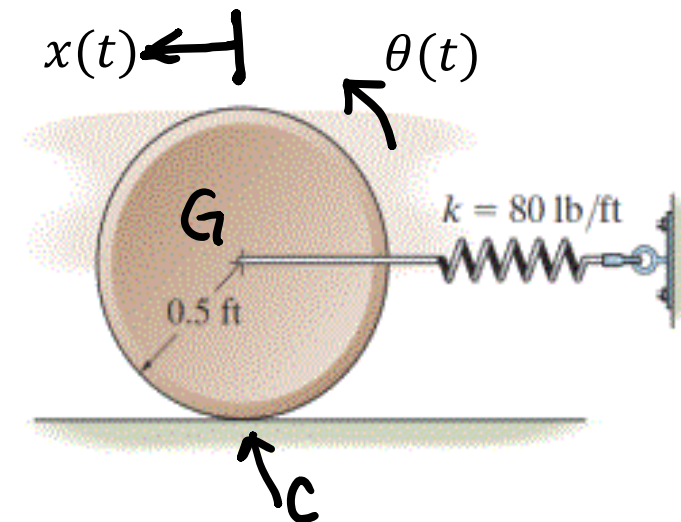
**Solution:** Use the coordinate system as shown. Imposing the no-slip kinematics at the contact point, we have  $x = r\theta$ . Apply conservation of energy principle. The PE and KE are:

$$V = \frac{1}{2}k(r\theta)^2, T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}I_G\dot{\theta}^2 = \frac{1}{2}I_C\dot{\theta}^2$$

where  $I_G = \frac{mr^2}{2}$ . Hence,  $I_C = \frac{3mr^2}{2}$ . Total energy is conserved,

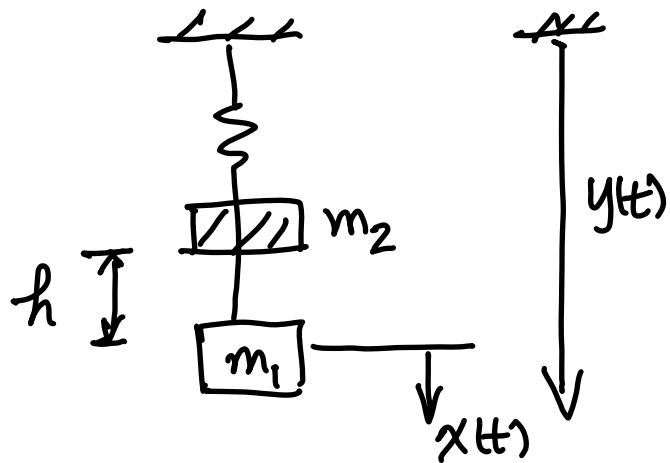
$$E = T + V = T_0 + V_0 = \frac{1}{2}I_C\dot{\theta}_{max}^2$$

where subscript 0 denotes initial values.



## In-class exercise:

A mass  $m_1$  is in static equilibrium position when a mass  $m_2$  is dropped a height of  $h$  and impacts  $m_1$ . After impact, there is no rebound (i.e., the coefficient of restitution  $e=0$ ). Determine the subsequent vibratory motion.

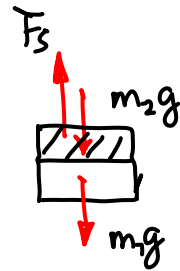


Key concept (key physics) :

After impact, both masses will move together initially. The velocity after impact defines the initial velocity of the system of masses. This goes into initial condition of the ODE.

# In-class exercise:

Before deriving the suitable equation of motion, let's understand the basic physics and use of coordinates. Let  $x(t)$  be the displacement from the equilibrium position of  $m_1$  alone,  $y(t)$  be the displacement measured from a fixed position, i.e., this is the absolute displacement. The coordinate  $y(t)$  may also be viewed as measured from the unstretched position of the spring.



Free-body diagram (best to base on absolute coordinates in which Newton's Law can be applied safely)

Apply Newton's law, we obtain

$$(m_1 + m_2)\ddot{y} = m_1g + m_2g - \cancel{F_s} \rightarrow k y \quad (1)$$

Let  $x = y - \frac{m_1g}{k}$  i.e.,  $x(t)$  is the coordinate relative to the eqn position of  $m_1$  alone

$$\rightarrow \ddot{x} = \ddot{y}$$

$$\begin{aligned} \text{From Eq (1)} \Rightarrow (m_1 + m_2)\ddot{x} &= \cancel{m_1g} + m_2g - k(x + \cancel{\frac{m_1g}{k}}) \\ \Rightarrow (m_1 + m_2)\ddot{x} + kx &= m_2g \end{aligned} \quad (2)$$

## In-class exercise:

Applying impulse-momentum principle to the impact of the system, we have conservation of the linear momentum of the system along the impact direction.

Velocity of  $m_2$  right before impact is obtained by energy conservation of the drop  $\Rightarrow \frac{1}{2} m_2 (v_2')^2 = m_2 g h \Rightarrow v_2' = \sqrt{2gh}$

$\rightarrow m_2 v_2' = (m_1 + m_2) v \Rightarrow v = \frac{m_2}{m_1 + m_2} \sqrt{2gh}$  (velocity of masses right after impact)

The initial conditions for eqn. (2) are

$$x(0) = 0 \quad \text{and} \quad \dot{x}(0) = \frac{m_2}{m_1 + m_2} \sqrt{2gh} \quad (3)$$

Equations (2) and (3) define the governing eqn of motion and initial conditions.

## In-class exercise:

We can also rewrite eqn (1) by "removing" all the  $mg$  terms by defining  $\tilde{x} = y - \underbrace{\left(\frac{m_1 g}{k} + \frac{m_2 g}{k}\right)}_{\text{eqn of the system}}$

The governing eqn of motion becomes

$$(m_1 + m_2)\ddot{\tilde{x}} + k\tilde{x} = 0 \quad (4)$$

and the initial conditions are

$$\tilde{x}(0) = -\frac{m_2 g}{k} \quad \text{and} \quad \dot{\tilde{x}}(0) = \frac{m_2}{m_1 + m_2} \sqrt{2gh} \quad (5)$$

Solving eqns. (2) and (3) is the same as solving eqns. (4) and (5). The advantage of the latter formulation is that we would be solving a homogeneous eqn rather than a non-homogeneous eqn of (2). The relation between  $x(t)$  and  $\tilde{x}(t)$  are

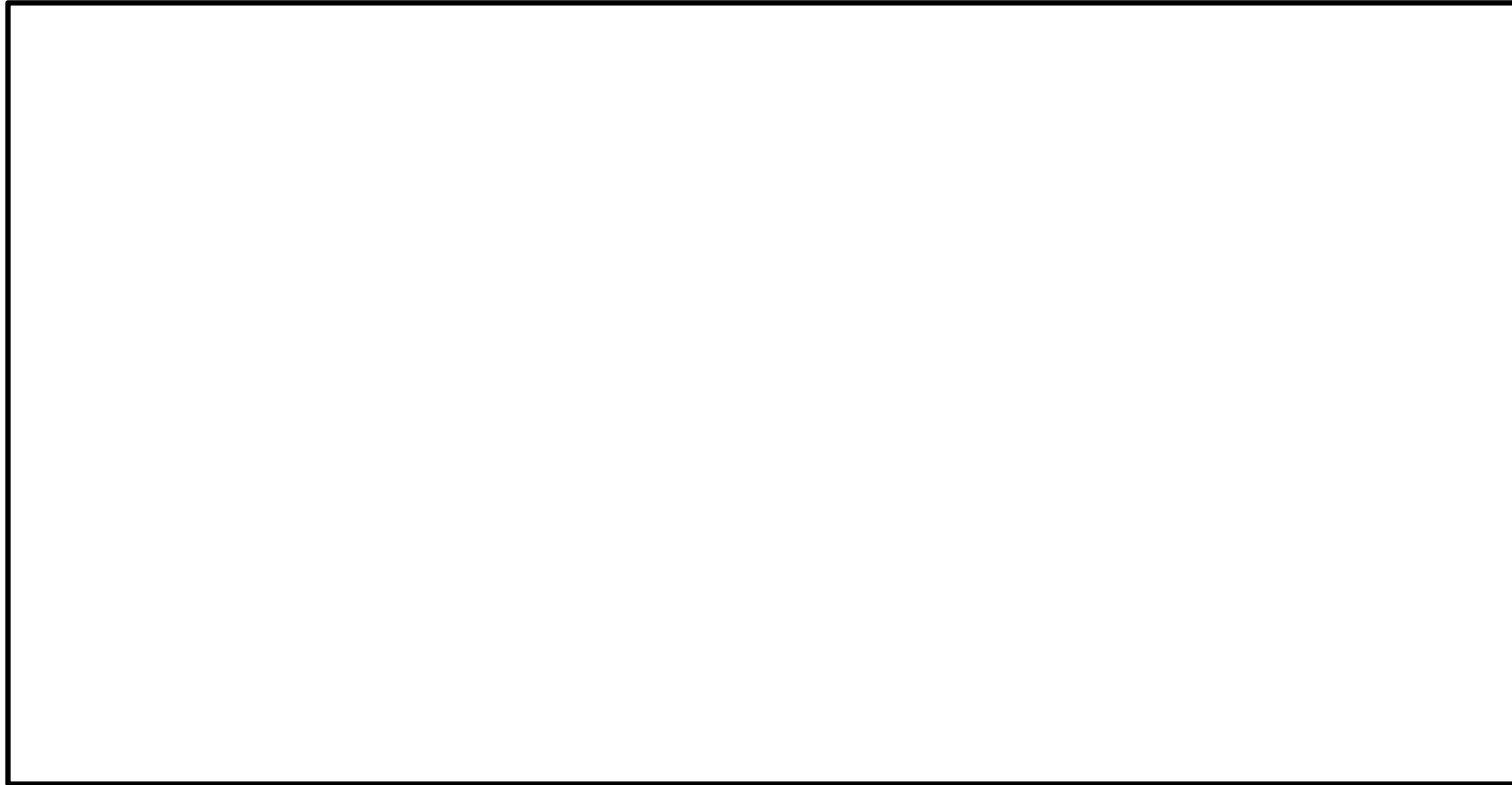
$$\tilde{x}(t) = x(t) - \frac{m_2 g}{k} \quad (6)$$

## In-class exercise:

Solution of Eqns. (2) and (3) as forced response based on  $x(t)$  coordinate.

Solution of Eqns (2) and (3) :

$$x(t) = \underbrace{x_h(t)}_{\text{homogeneous soln}} + \underbrace{x_p(t)}_{\text{particular soln}} \quad (7)$$



## In-class exercise:

Solution of Eqns. (4) and (5) as free response based on  $\tilde{x}(t)$  coordinate.

Solution of eqns (4) and (5)

$$\tilde{x}(t) = \tilde{A} \cos \omega_n t + \tilde{B} \sin \omega_n t$$

$$\tilde{x}(0) = -\frac{m_2 g}{k} \Rightarrow \tilde{A} = -m_2 g/k$$

$$\dot{\tilde{x}}(0) = v_0 \Rightarrow \tilde{B} \omega_n = v_0 \Rightarrow \tilde{B} = v_0/\omega_n \quad (\tilde{B} = B)$$

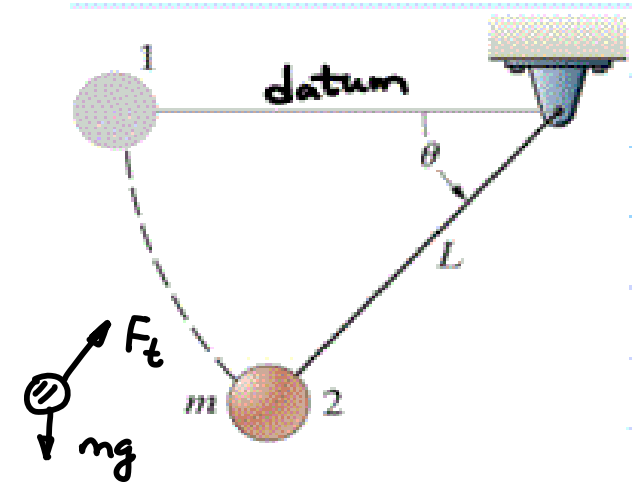
$$\text{Thus, } \tilde{x}(t) = -\frac{m_2 g}{k} \cos \omega_n t + m_2 \sqrt{\frac{2gh}{k(m_1+m_2)}} \sin \omega_n t \quad (9)$$

Since  $\tilde{x}(t) = x(t) - \frac{m_2 g}{k} \Rightarrow$  Eqns (8) and (9) represent the same soln.

## A basic problem in vibration:

The ball is released from rest in position 1 and swings to position 2.

- Determine the velocity of the ball in position 2.
- Draw graphs of KE, PE and total energy as a function of  $\theta$ .
- Determine the tension of the string as a function of  $\theta$ .
- Determine the governing equation of motion.



(a) The tension  $F_t$  does not do work  
since it is normal to the path.

$$\Delta T = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 \underset{0}{=} = \frac{1}{2} m v_2^2$$

$$\Delta V = mg(y_2 - y_1) = mg(-L \sin \theta)$$

$$\Delta T + \Delta V = 0 \Rightarrow \frac{1}{2} m v_2^2 = mgL \sin \theta \Rightarrow v_2^2 = 2gL \sin \theta$$

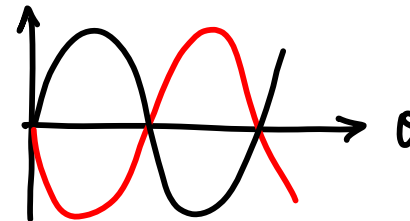
(same result as before)

(b)

$$\begin{cases} \text{Potential energy PE, } V = -mgL \sin \theta \\ \text{Kinetic energy KE, } T = \frac{1}{2} m v^2 = mgL \sin \theta \end{cases}$$

Total mechanical energy,  $E = T + V = 0$  (constant)

— KE  
— PE



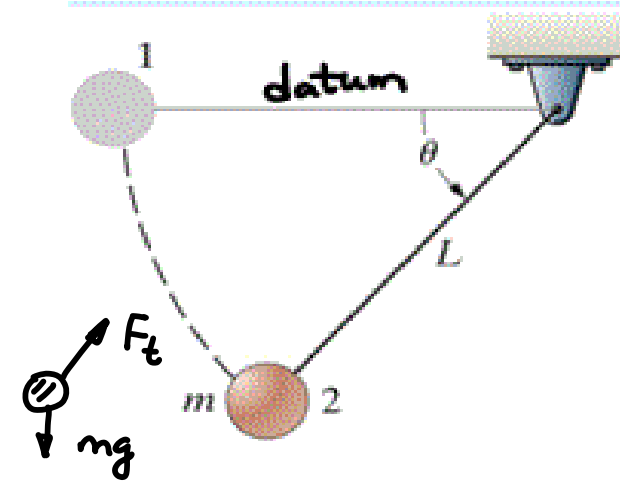
↑ This value is conserved (does not change over time) and it depends on the defn. of the PE.

## A Basic Problem:

Prob. 15.83

The ball is released from rest in position 1 and swings to position 2.

- (a) Determine the velocity of the ball in position 2.
- (b) Draw graphs of KE, PE and total energy as a function of  $\theta$ .
- (c) Determine the tension of the string as a function of  $\theta$ .
- (d) Determine the governing equation of motion.



$$\begin{aligned} \text{(c)} \quad F_t - mg \sin \theta &= m \frac{V^2}{\rho} \Rightarrow F_t = mg \sin \theta + m \frac{V^2}{L} \\ &= mg \sin \theta + m \left( \frac{2gL \sin \theta}{L} \right) \\ &= 3mg \sin \theta \end{aligned}$$

$$\text{when } \theta = 90^\circ, F_t = 3mg$$

$$\text{(d)} \quad mg \cos \theta = m a_t \Rightarrow g \cos \theta = a_t \quad \leftarrow \text{This equation determines the rate of change of the velocity } v.$$

## A basic problem of vibration of a simple pendulum:

Consider a simple pendulum released from rest when the pendulum is at the horizontal position. We would like to study this problem to understand how to formulate the equation of motion.

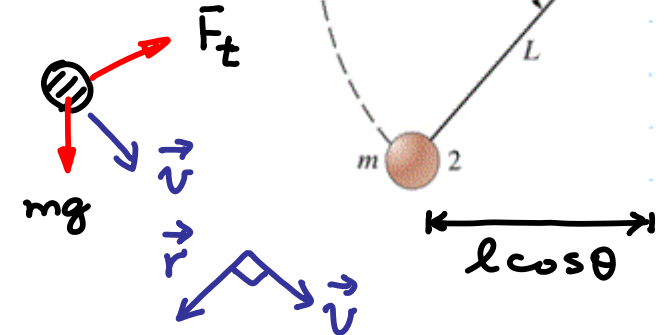
Eqn of motion:

$$\Sigma M_A = \frac{d}{dt}(\vec{r} \times m\vec{v}) \cdot \hat{k}$$

$$\Rightarrow mgl \cos \theta = \frac{d}{dt}[(l)(ml\dot{\theta})]$$

$$\Rightarrow ml^2\ddot{\theta} - mgl \cos \theta = 0$$

FBD:



Conservative system:

$$\Delta T + \Delta V = \underbrace{U_{nc}}_{\text{zero for this problem because } F_t \text{ is } \perp \text{ to trajectory}}, \quad dU = \vec{F} \cdot d\vec{r} \Rightarrow U = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$

$$\Delta T = T_2 - T_1 = \frac{1}{2}mv^2 - 0; \quad \Delta V = V_2 - V_1 = -mgl \sin \theta - 0$$

Energy conservation means that  $E = T + V = \text{constant}$  ( $E = \text{total mechanical energy}$ )

$$\Rightarrow \frac{dE}{dt} = 0 \Rightarrow \frac{d}{dt}(\frac{1}{2}mv^2 - mgl \sin \theta) = 0, \quad v = l\dot{\theta}$$

$$\Rightarrow ml^2\ddot{\theta} - (mgl \cos \theta)\dot{\theta}$$

$$\Rightarrow \underline{(ml^2\ddot{\theta} - mgl \cos \theta)\dot{\theta} = 0}$$

this is the eqn of motion

Now suppose the simple pendulum is initially at rest in the vertical position. Determine the governing equation motion for small amplitude of oscillation and the natural frequency of oscillation (concept for linear systems).

$$T = \frac{1}{2}mv^2 = \frac{1}{2}ml^2\dot{\theta}^2, \quad V = -mgl\cos\theta$$

$$\frac{dE}{dt} = 0 \Rightarrow \frac{d}{dt} \left( \frac{1}{2}ml^2\dot{\theta}^2 - mgl\cos\theta \right) = 0$$

$$\Rightarrow ml^2\ddot{\theta}\dot{\theta} + mgl\sin\theta\dot{\theta} = 0$$

$$\Rightarrow ml^2\ddot{\theta} + mgl\sin\theta = 0 \quad (\text{eqn of motion})$$

$$\text{For small angle, } \sin\theta \approx \theta \Rightarrow ml^2\ddot{\theta} + mgl\theta = 0 \quad (\text{linear ODE})$$

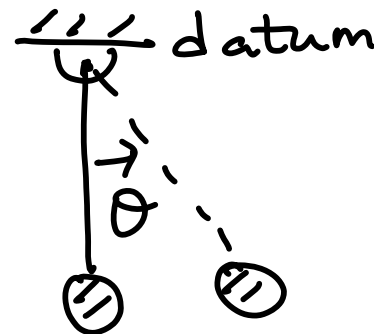
$$ml^2\ddot{\theta} + mgl\theta = 0 \Rightarrow \ddot{\theta} + \frac{g}{l}\theta = 0 \quad (\ddot{\theta} = \frac{d^2\theta}{dt^2})$$

General solution of this vibratory system is:

$$\theta(t) = A \sin \omega_0 t + B \cos \omega_0 t; \quad \omega_0 = \sqrt{\frac{g}{l}}$$

$\nwarrow$  determined from initial conditions

$=$  natural frequency



Natural frequency is the fundamental frequency that the pendulum will oscillate.

Solve numerically:

Define state variables  $\vec{u} = \begin{Bmatrix} \theta \\ \dot{\theta} \end{Bmatrix}$

$$\frac{d\vec{u}}{dt} = \begin{Bmatrix} \dot{\theta} \\ \ddot{\theta} \end{Bmatrix} = \begin{Bmatrix} u_2 \\ -\frac{g}{l} \sin u_1 \end{Bmatrix}$$

$$m l^2 \ddot{\theta} + m g l \sin \theta = 0 \\ \Rightarrow \ddot{\theta} = -\frac{g}{l} \sin \theta$$

function udot = dudt (t, u)

udot = [ u(2); -g/l \* sin(u(1)) ];

[t, u] = ode23 (@dudt, tspan, u0);

output:

$$u(:, 1) = \theta(t)$$

$$u(:, 2) = \dot{\theta}(t)$$

The total mechanical energy  $E = T + V = \text{constant} = \text{initial energy}$   
 $\frac{1}{2} m (l \dot{\theta})^2 + (-m g l \cos \theta) = \text{const} = E_0$  (depends on initial conditions)

$$\dot{\theta}^2 = \frac{2(E_0 + m g l \cos \theta)}{m l^2} \Rightarrow \dot{\theta} = \sqrt{\frac{2(E_0 + m g l \cos \theta)}{m l^2}} = f(\theta)$$

Since  $\dot{\theta} = \frac{d\theta}{dt}$ , we can use this kinematic relation to obtain

$$\int \frac{d\theta}{f(\theta)} = \int dt$$

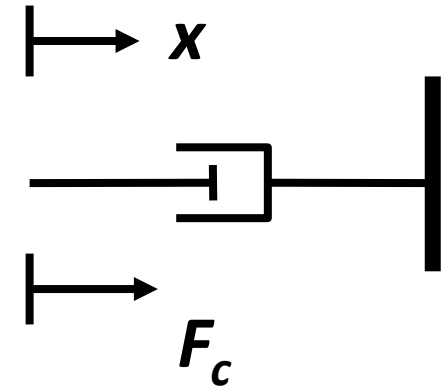
Numerically integrate the above to obtain the period of oscillation.

# Viscous Damping Model

All real systems dissipate energy when they vibrate. To account for this we must consider damping. The simplest form of damping (from a mathematical viewpoint) is called viscous damping. A (linear) viscous damper (or **dashpot**) produces a force that is proportional to velocity. Note that damping can only be measured using dynamic tests (recall that stiffness can be measured by static tests).

The force in the dashpot is proportional to the damping constant  $c$ :

$$F_c = -cv(t) = -c\dot{x}(t)$$



# Governing equation for free vibration of damped systems

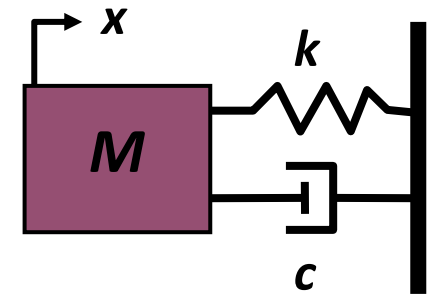
For this damped SDOF system, the force acting on the mass is due to the spring and the dashpot

$$m\ddot{x}(t) = -kx - c\dot{x}(t)$$

or,

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = 0$$

Displacement



To solve the above linear ordinary differential equation (ODE) with constant coefficients, assume a solution of the following complex form (this is the method of separation of variables):

$$x(t) = ae^{\lambda t}$$

# Solution for free vibration of damped systems

The velocity and acceleration can then be written as:

$$\dot{x}(t) = \lambda a e^{\lambda t}, \quad \ddot{x}(t) = \lambda^2 a e^{\lambda t}$$

Substituting these expressions in the governing equation gives:

$$a e^{\lambda t} (m \lambda^2 + c \lambda + k) = 0$$

For non-trivial solutions, we require

$$a e^{\lambda t} \neq 0 \Rightarrow \lambda^2 + \frac{c \lambda}{m} + \omega_n^2 = 0 \text{ or } \lambda^2 + 2\zeta \omega_n \lambda + \omega_n^2 = 0$$

The above is the characteristic equation or frequency equation whose solution gives the characteristic values or eigenvalues of the system.

The eigenvalues define the vibration characteristics of the system.

# Solution for free vibration of damped systems (cont.)

Recall that the damping ratio is:

$$\zeta = \frac{c}{2\sqrt{km}}$$

The characteristic equation then becomes:

$$\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 = 0$$

Solving the above equation gives the characteristics values or eigenvalues

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \quad \text{or} \quad \lambda_{1,2} = \underbrace{-\zeta\omega_n}_{\text{decaying soln.}} \pm \underbrace{i\omega_n\sqrt{1 - \zeta^2}}_{\text{oscillatory solution}}$$

**Three types of motions** are possible depending on the values of  $\zeta$ . Note that the ODE is often written as:

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2 x(t) = 0$$

# Concept of Damping Ratio

- **Three** key variables and terminologies for damped SDOF systems:

- $\omega_n = \sqrt{\frac{k}{m}}$ , natural frequency (rad/s); the frequency at which the undamped system vibrates
- $\zeta \omega_n = \frac{c}{2m}$ , where  $c$  is the damping constant and  $\zeta$  is called the damping ratio;  
 $\zeta = \frac{c}{2\sqrt{mk}}$  (a non-dimensional parameter)

- $\omega_d = \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} = \omega_n \sqrt{1 - \zeta^2}$ , damped natural frequency (rad/s)

Hence the damped period is:  $T_d = 2\pi / \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2}$ .

For lightly damped systems,  $\omega_d \cong \omega_n$

# Concept of Damping Ratio

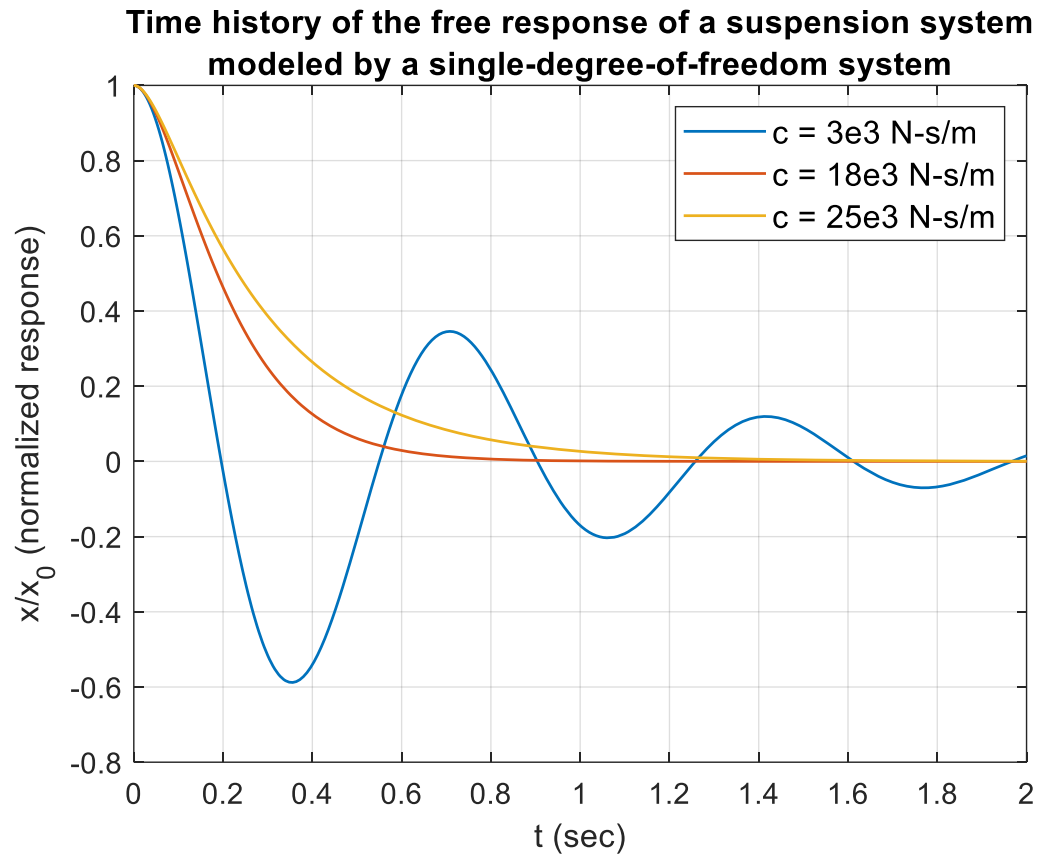
- Work with the normalised governing equation of motion (divide the general equation by  $m$ ):

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = 0, \text{ initial conditions: } x(0) = x_0, \dot{x}(0) = v_0$$

- When the damping is small, i.e.,  $\zeta < 1$  and  $\frac{k}{m} - \left(\frac{c}{2m}\right)^2 > 0$ , the motion of the suspension system is said to be underdamped.
- The critically damped and overdamped vibrations correspond to  $\zeta = 1$  or  $\frac{k}{m} - \left(\frac{c}{2m}\right)^2 = 0$ , and  $\frac{k}{m} - \left(\frac{c}{2m}\right)^2 < 0$  or  $\zeta > 1$ , respectively.
- The damping ratio determines the characteristics of the response, not the actual value of the damping constant.

# Response under different damping ratios

- Sample MATLAB simulation. Choose  $m = 1,000$  kg,  $k = 81,000$  N/m,  $v_0 = 0$ .



Motions exhibit energy dissipation:

$\zeta < 1$ , oscillatory

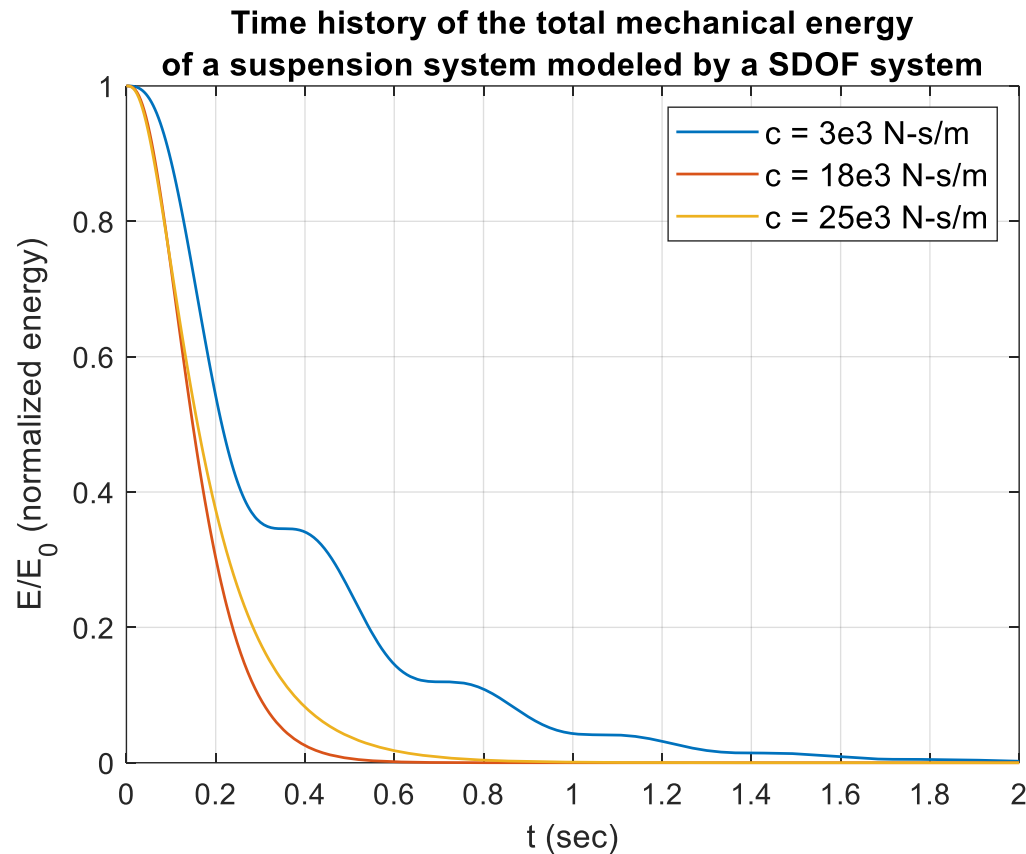
$\zeta \geq 1$ , purely decaying

$c = 18,000$  N-s/m is the critically damped case for this example.

A discussion of the mathematical and simulated solutions of the response will be presented later.

# Energy dissipation under different damping ratios

- Sample MATLAB simulation. Choose  $m = 1,000$  kg,  $k = 81,000$  N/m,  $v_0 = 0$ .



Energy dissipates uniformly over time. Critically damped case seems to dissipate energy most optimally. For optimisation problem, examine an energy functional called action:

$$action = \int_0^{t^*} E(t) dt$$

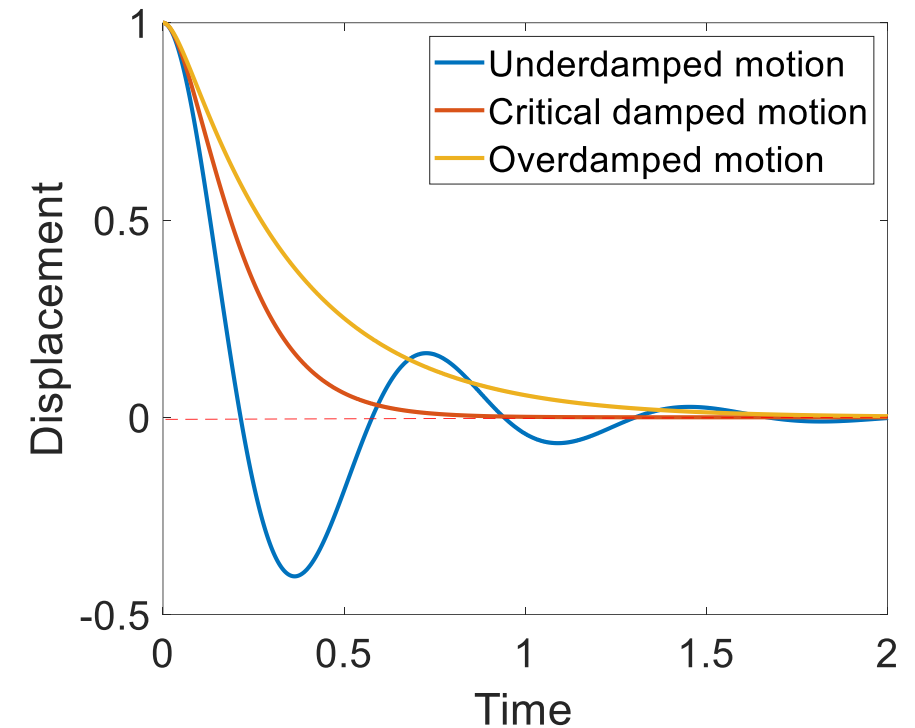
where,  $t^*$  is a user-defined design parameter.

$c = 18,000$  N-s/m is the critically damped case for this example.

# Solution of free vibration of damped systems

Three types of motions are possible depending on the values of  $\zeta$ .

| Values of $\zeta$ | Types of motions         | Characteristics of motion                                                                |
|-------------------|--------------------------|------------------------------------------------------------------------------------------|
| $\zeta < 1$       | Underdamped motion       | The mass oscillates with decay and returns to its rest position as energy is dissipated. |
| $\zeta = 1$       | Critically damped motion | No oscillation. This is a motion between the underdamped and overdamped motions.         |
| $\zeta > 1$       | Overdamped motion        | The mass returns to its rest position without any oscillation.                           |



# Underdamped free vibration

An underdamped motion occurs when  $\zeta < 1$ . The roots of the characteristic equation are complex conjugate pairs

$$\lambda_{1,2} = -\zeta\omega_n \pm i\omega_n\sqrt{1 - \zeta^2}$$

The response is given by:

$$x(t) = e^{-\zeta\omega_n t} (a_1 e^{i\omega_d t} + a_2 e^{-i\omega_d t})$$

where,  $\omega_d = \omega_n\sqrt{1 - \zeta^2}$  is the damped natural frequency. The motion is oscillatory with exponential decay. This is the most common case in engineering. The constants  $a_1$  and  $a_2$  are determined from the initial conditions of the motion, which are needed to solve the ODE.

# Underdamped free vibration

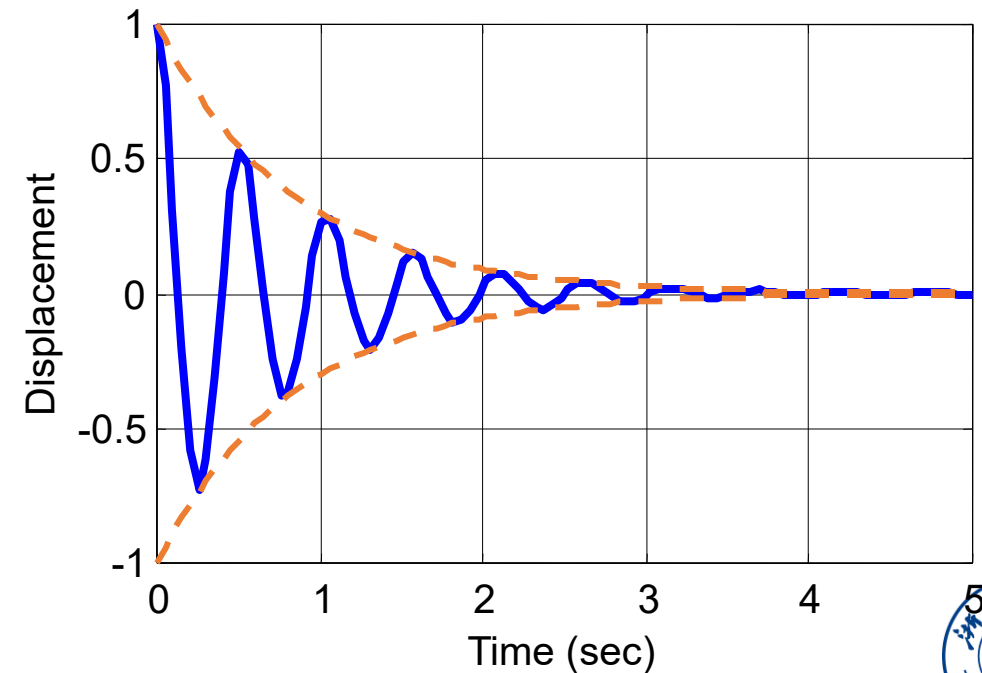
The response is a real function and it is more commonly expressed as:

$$x(t) = X_0 e^{-\zeta\omega_n t} \cos(\omega_d t - \phi_0)$$

where the amplitude  $X$  and the phase  $\phi$  are determined by invoking the initial conditions.

$$X_0 = \frac{1}{\omega_d} \sqrt{(x_0 \omega_d)^2 + (v_0 + \zeta \omega_n x_0)^2}$$

$$\phi_0 = \tan^{-1} \left( \frac{v_0 + \zeta \omega_n x_0}{x_0 \omega_d} \right)$$



# Underdamped free vibration

Alternatively, response in terms of the sine function is expressed as:

$$x(t) = X_0 e^{-\zeta \omega_n t} \sin(\omega_d t + \psi_0)$$

where the amplitude  $X_0$  and the phase  $\psi_0$  are again determined by invoking the initial conditions.

$$X_0 = \frac{1}{\omega_d} \sqrt{(x_0 \omega_d)^2 + (v_0 + \zeta \omega_n x_0)^2}$$

$$\psi_0 = \tan^{-1} \left( \frac{x_0 \omega_d}{v_0 + \zeta \omega_n x_0} \right)$$

Again, we have  $(\tan \phi_0)(\tan \psi_0) = 1$ , hence  $\phi_0 + \psi_0 = \pi/2$ .

# Characteristics of underdamped free vibration

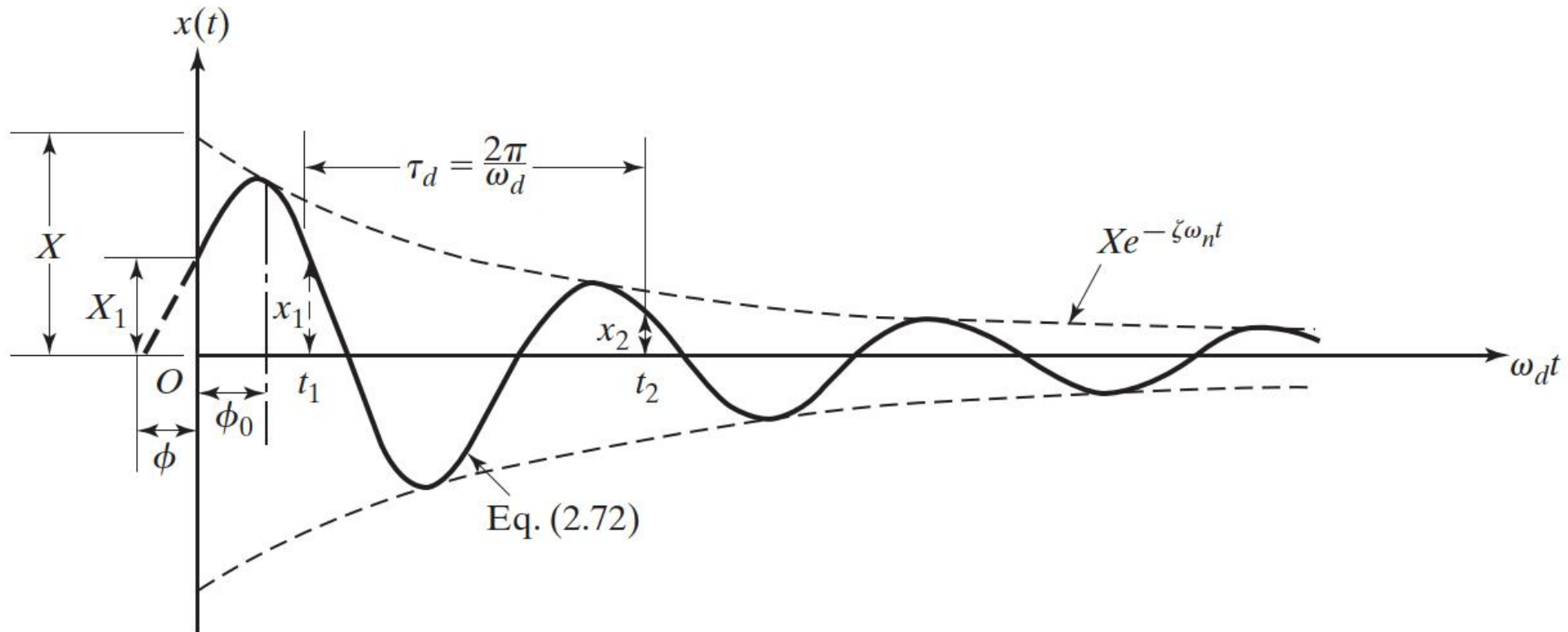


FIGURE 2.22 Underdamped solution.

Rao, S. S., *Mechanical Vibrations*, 6th edition.

# Critically damped free vibration

Critically damping occurs when  $\zeta = 1$ . The damping coefficient in this case is equal to

$$2\zeta\omega_n = \frac{c}{m} \Rightarrow c_{cr} = 2m\omega_n = 2\sqrt{km}$$

The roots of the characteristic equation are repeated

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} = -\omega_n$$

To seek the second independent solution, consider  $\frac{d}{d\lambda}(e^{\lambda t}) = te^{\lambda t}$ .

The response is thus given by:

$$x(t) = a_1 e^{-\omega_n t} + a_2 t e^{-\omega_n t} = (a_1 + a_2 t) e^{-\omega_n t}$$

# Critically damped free vibration

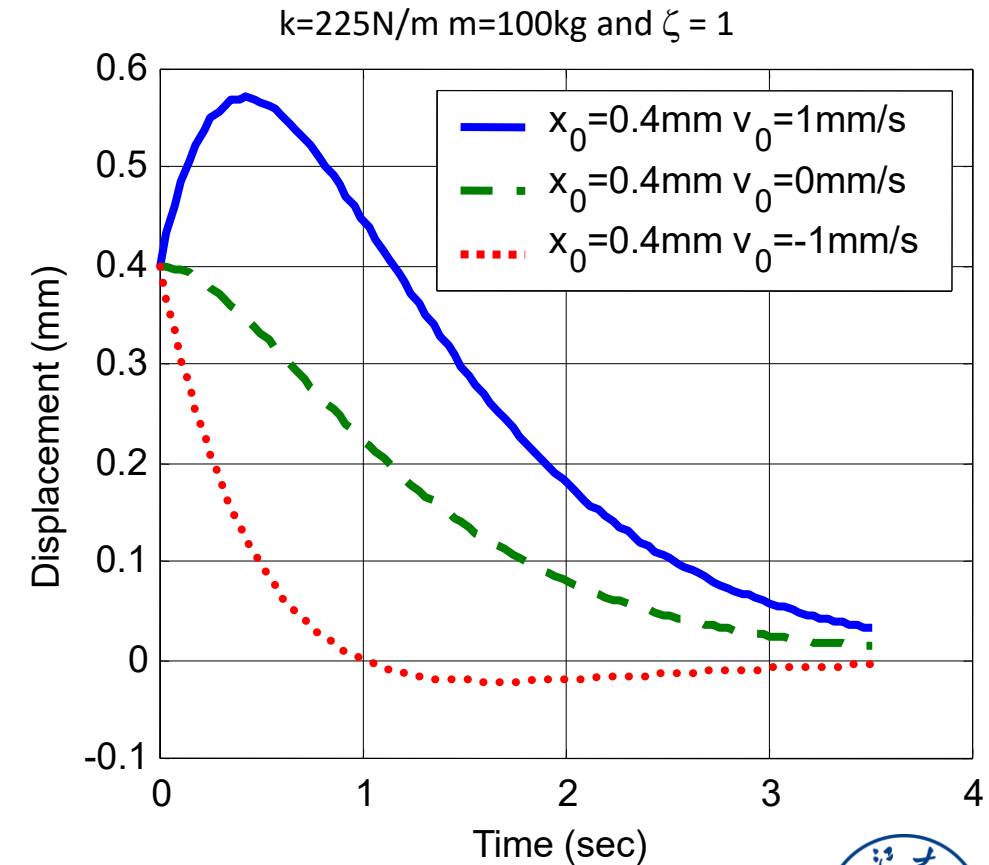
The constants  $a_1$  and  $a_2$  can be determined by invoking the initial conditions

$$x(t) = (a_1 + a_2 t)e^{-\omega_n t} \Rightarrow a_1 = x_0$$

$$v(t) = \dot{x}(t) = (-\omega_n a_1 - \omega_n a_2 t + a_2)e^{-\omega_n t}$$

$$v_0 = -\omega_n a_1 + a_2 \Rightarrow a_2 = v_0 + x_0 \omega_n$$

There is no oscillation. Useful for design of door mechanisms, analog gauges, etc.



# Overdamped free vibration

An overdamped case occurs when  $\zeta > 1$ . The eigenvalues are

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

The response is:

$$x(t) = e^{-\zeta\omega_n t} \left( a_1 e^{\omega_n t \sqrt{\zeta^2 - 1}} + a_2 e^{-\omega_n t \sqrt{\zeta^2 - 1}} \right)$$

The constants are determined from the initial conditions:

$$a_1 = \frac{v_0 - \left( -\zeta - \sqrt{\zeta^2 - 1} \right) \omega_n x_0}{2\omega_n \sqrt{\zeta^2 - 1}}, \quad a_2 = \frac{-v_0 + \left( -\zeta + \sqrt{\zeta^2 - 1} \right) \omega_n x_0}{2\omega_n \sqrt{\zeta^2 - 1}}$$

# Overdamped free vibration – Derivation

The eigenvalues are:  $\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$  (choose  $\lambda_1$  as the plus sign)

The general solution is:  $x(t) = a_1 e^{\lambda_1 t} + a_2 e^{\lambda_2 t}$

The constants are determined from the initial conditions:

$$x_0 = a_1 + a_2$$

The velocity is expressed as:

$$\dot{x} = a_1 \lambda_1 e^{\lambda_1 t} + a_2 \lambda_2 e^{\lambda_2 t}$$

Thus, we have

$$v_0 = a_1 \lambda_1 + a_2 \lambda_2$$

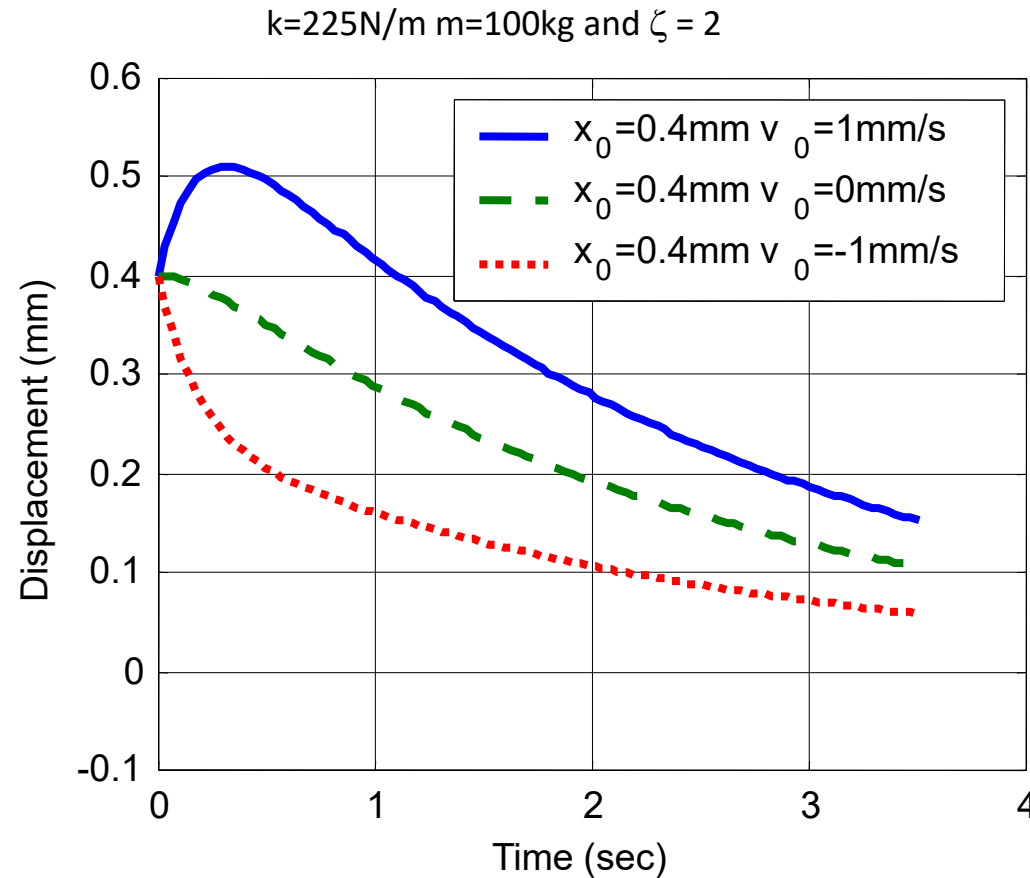
Formulate a matrix equation:  $\begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} x_0 \\ v_0 \end{Bmatrix} \Rightarrow \begin{Bmatrix} a_1 \\ a_2 \end{Bmatrix} = \frac{\begin{bmatrix} -\lambda_2 & 1 \\ \lambda_1 & -1 \end{bmatrix}}{(\lambda_1 - \lambda_2)} \begin{Bmatrix} x_0 \\ v_0 \end{Bmatrix}$

Solving the above matrix equation gives the answer.

$$a_1 = \frac{-\lambda_2 x_0 + v_0}{2\omega_n \sqrt{\zeta^2 - 1}}, \quad a_2 = \frac{\lambda_1 x_0 - v_0}{2\omega_n \sqrt{\zeta^2 - 1}}$$

# Characteristics of overdamped free vibration

The response is slower than the critically damped case



## Summary of Solutions for Free Response:

$$x(t) = e^{-\zeta\omega_n t} \left( x_0 \cos \omega_d t + \frac{v_0 + \zeta\omega_n x_0}{\omega_d} \sin \omega_d t \right), \quad \text{for } \zeta \neq 1$$

$$x(t) = x_0 e^{-\omega_n t} + (v_0 + \omega_n x_0) t e^{-\omega_n t}, \quad \text{for } \zeta = 1$$

**Note:** Although the above expression for  $\zeta > 1$  is correct, it is more customary to express in term of the decaying exponential functions to reflect the actual motion.

$$x(t) = e^{-\zeta\omega_n t} \left( a_1 e^{\omega_n t \sqrt{\zeta^2 - 1}} + a_2 e^{-\omega_n t \sqrt{\zeta^2 - 1}} \right)$$

$$a_1 = \frac{v_0 - \left( -\zeta - \sqrt{\zeta^2 - 1} \right) \omega_n x_0}{2\omega_n \sqrt{\zeta^2 - 1}}, \quad a_2 = \frac{-v_0 + \left( -\zeta + \sqrt{\zeta^2 - 1} \right) \omega_n x_0}{2\omega_n \sqrt{\zeta^2 - 1}}$$

## Summary of Solutions for Free Response:

Above formulas in terms of the system parameters are:

$$x(t) = e^{-ct/2m} \left\{ x_0 \cos \left[ \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} t \right] + \frac{v_0 + x_0(c/2m)}{\sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2}} \sin \left[ \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} t \right] \right\}, \quad \text{for } \zeta \neq 1$$

$$x(t) = x_0 e^{-\sqrt{k/m} t} + \left( v_0 + x_0 \sqrt{k/m} \right) t e^{-\sqrt{k/m} t}, \quad \text{for } \zeta = 1$$

These expressions are very complicated. Hence, in computer programming, one should always first define the three key variables,  $\omega_n$ ,  $\zeta$ , and  $\omega_d$ . Then use the formulas on the previous page to obtain the response solutions.

# Summary of Solutions for Free Response:

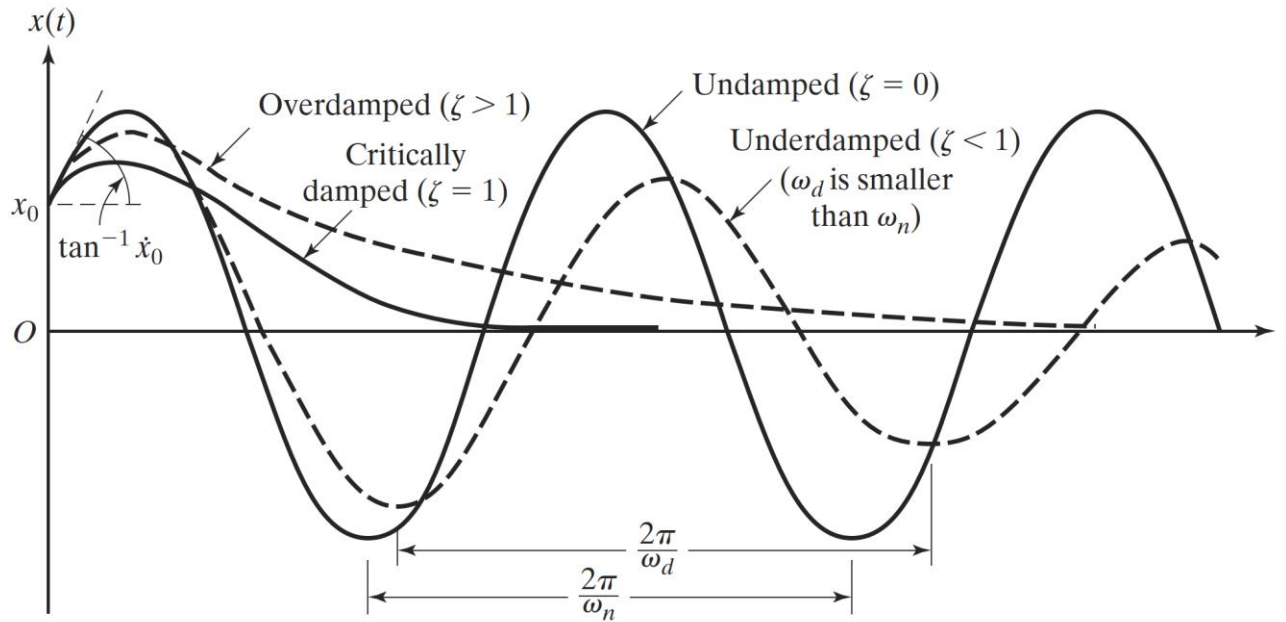


FIGURE 2.24 Comparison of motions with different types of damping.

Source: Rao, *Mechanical Vibrations*, 6e.

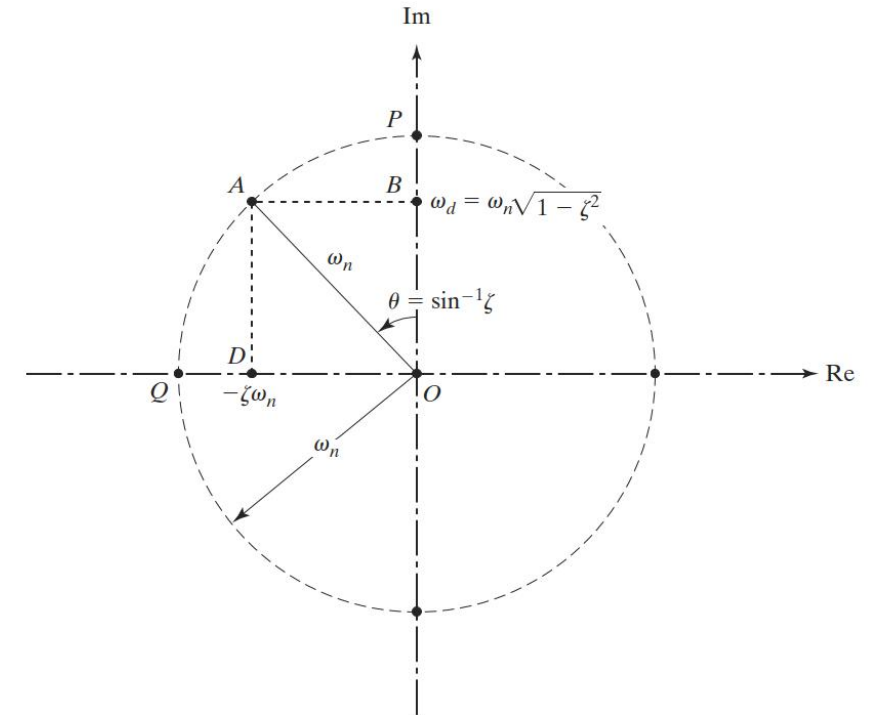


FIGURE 2.33 Interpretations of  $\omega_n$ ,  $\omega_d$ , and  $\zeta$ .

# Summary of free vibration of a damped SDOF system

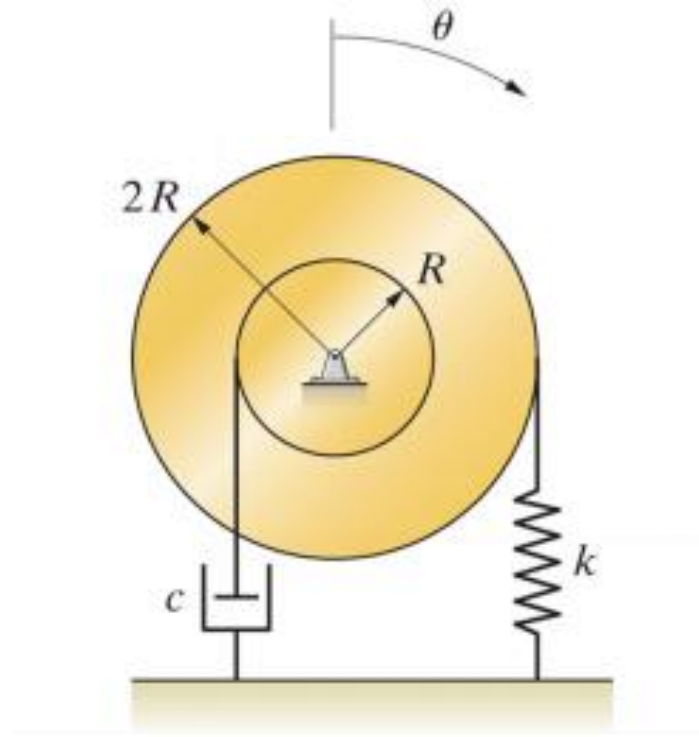
## Natural Characteristics of a Damped Oscillator

| Damping ratio | Level of damping     | Oscillatory response | Stability                                                                                                              | Speed of response         | Time of constant                                           |
|---------------|----------------------|----------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------|
| $\zeta < 1$   | Underdamped          | Yes                  | Asymptotically stable<br>(less stable than $\zeta = 1$ case<br>but not necessarily less<br>stable than the overdamped) | Better than<br>overdamped | $\frac{1}{\zeta \omega_n}$                                 |
| $\zeta > 1$   | Overdamped           | No                   | Asymptotically stable;<br>Less stable than the critically<br>damped case                                               | Lower than<br>critical    | $\frac{1}{\zeta \omega_n \pm \omega_n \sqrt{\zeta^2 - 1}}$ |
| $\zeta = 1$   | Critically<br>damped | No                   | Asymptotically stable;<br>Most stable                                                                                  | Good                      | $\frac{1}{\omega_n}$                                       |

Source: De Silva, *Fundamentals of Vibration*

## Example 1

Consider the following vibratory system with  $R = 250$  mm,  $k = 150$  N/m, and the moment of inertia of the disk about its center is  $I = 2$  kg–m<sup>2</sup>.



# Example 1

- a) Derive the governing equation of motion.
- b) At what value of  $c$  is the system critically damped?
- c) At  $t = 0$ , the spring is unstretched and the clockwise angular velocity of the disk is 10 rad/s. If the system is critically damped, determine the response  $q(t)$  as a function of time. Plot this response graphically.
- d) Determine the maximum resulting angular displacement of the disk and the time at which it occurs. You can do this numerically if you prefer.
- e) Determine (numerically or graphically) the time of the first occurrence for the system to reduce its response to 10% of the maximum amplitude.
- f) Change one physical parameter and repeat part (e) to understand the concept of parametric design and optimization commonly employed in the industry.

# Solution of Example 1

a) Derive the governing equation of motion

**Solution:** The sum of the moments about the center of the stepped disk is

$$\sum M = -R \left( R c \frac{d\theta}{dt} \right) - 2R(2Rk\theta).$$

From the equation of angular motion

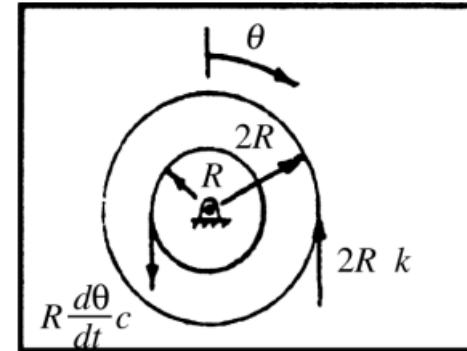
$$\sum M = I \frac{d^2\theta}{dt^2}.$$

The equation of motion is

$$I \frac{d^2\theta}{dt^2} + R^2 c \frac{d\theta}{dt} + 4R^2 k \theta = 0.$$

The canonical form is

$$\boxed{\frac{d^2\theta}{dt^2} + 2d \frac{d\theta}{dt} + \omega^2 \theta = 0}, \text{ where } d = \frac{R^2 c}{2I}, \omega^2 = \frac{4R^2 k}{I}.$$



# Solution of Example 1

- (b) For critical damping,  $d^2 = \omega^2$ , from which

$$c = \frac{4}{R} \sqrt{kI} = 277 \text{ N-s/m}, \text{ and } d = 4.330 \text{ rad/s}.$$

- (c) The solution is of the form (see Eq. (21.25))  $\theta = Ce^{-dt} + Dte^{-dt}$ . Apply the initial conditions:  $\theta_0 = 0$ ,  $\dot{\theta}_0 = -10 \text{ rad/s}$ , from which  $C = 0$ , and  $D = -10$ . The solution is

$$\theta(t) = \dot{\theta}_0 t e^{-dt} = -10 t e^{-4.330t} \text{ rad/s}.$$

- (d) The maximum (or minimum) value of the angular displacement is obtained from

$$\frac{d\theta}{dt} = 0 = -10 e^{-4.330t} (1 - 4.33t) = 0,$$

from which the maximum/minimum occurs at

$$t_{\max} = \frac{1}{4.330} = 0.231 \text{ s}.$$

The angle is  $[\theta]_{t=t_{\max}} = -0.850 \text{ rad clockwise}$ .

Note that the notation:

$$d = \zeta \omega_n$$

Note that the spring is unstretched at equilibrium.

# Solution of Example 1

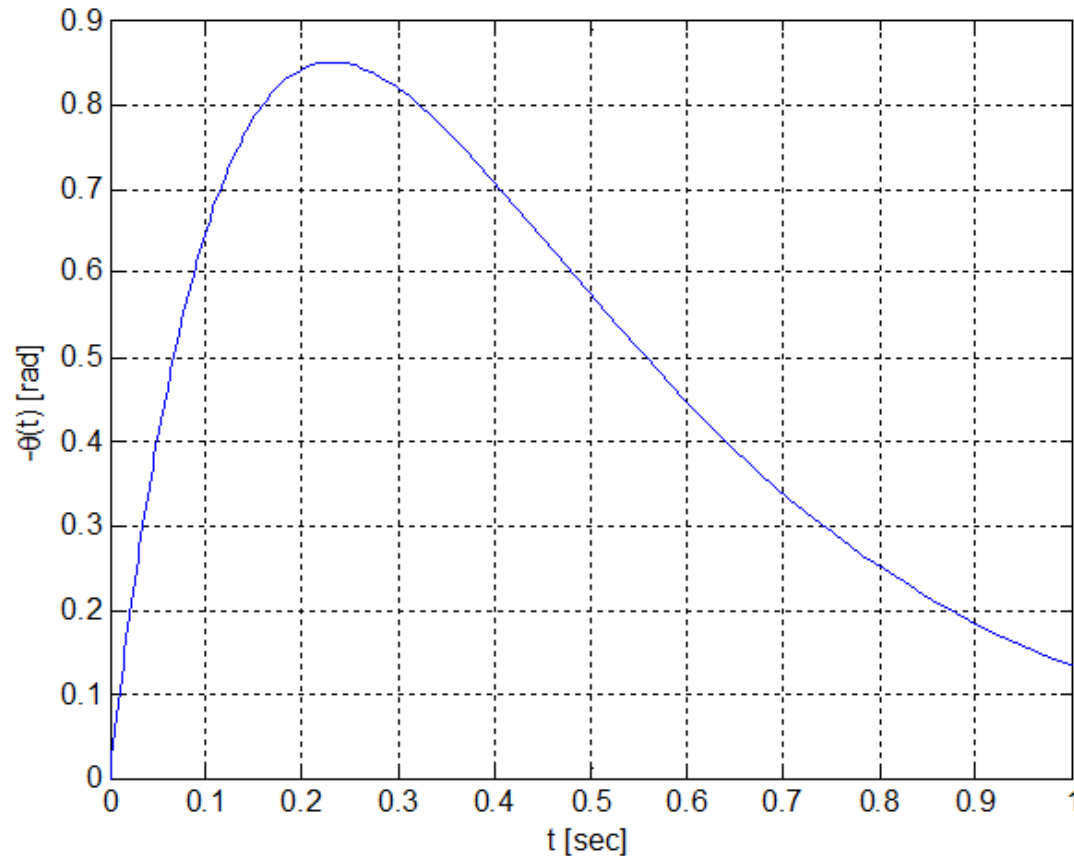
A time-history plot of the response is shown below.

Note that the counterclockwise angle (minus theta) is shown for better visualization.

One can use this plot to graphically find the value of the maximum response amplitude, though this value can easily be obtained analytically as shown above.

```
>> % Plot the counterclockwise angle (minus theta) for better visualization  
>> fplot(@(t) 10*t.*exp(-4.33*t),[0,1]), grid  
>> xlabel('t [sec]'), ylabel('-\theta(t) [rad]')
```

# Solution of Example 1



Graphical result: The maximum amplitude occurs at about 0.23 seconds with a value of 0.85 rad counterclockwise.

```
>> ginput(1)
```

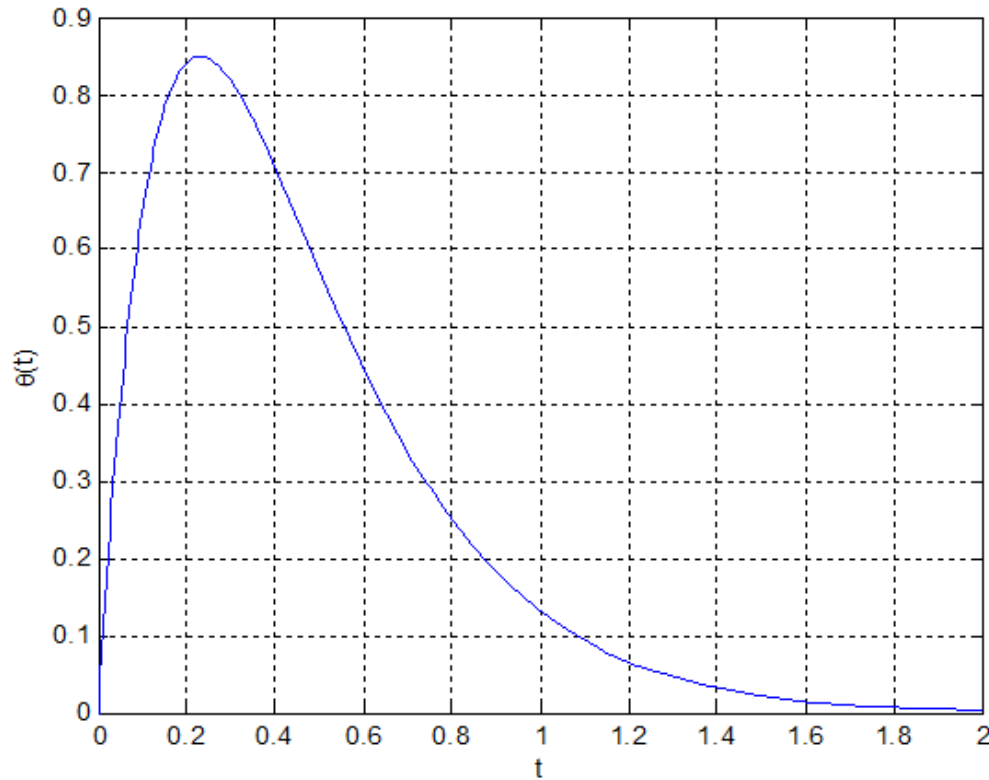
```
ans =  
    0.2278    0.8499
```

Using MATLAB fminsearch built-in function for a single variable: The maximum amplitude occurs at about 0.23 sec.

```
>> fminsearch(@(t) -10*t.*exp(-4.33*t),0.1)
```

```
ans =  
    0.2309
```

# Solution of Example 1



e) Solve  $10te^{-4.33t} = 0.085$ . It is quite impossible to solve analytically, so numerical solution must be sought. Using `fzero`, it is found to occur at about 1.13 sec.

```
>> fzero(@(t) 10*t.*exp(-4.33*t)-0.085,  
[0.2,2])
```

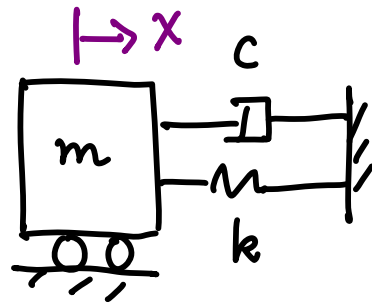
```
ans =  
    1.1291
```

f) When one physical parameter is changed, the system is probably not critically damped anymore. The equations will likely be more complicated to solve and numerical solutions should be sought.

# SDOF damped free vibration example:

For the case of zero initial velocity, apply compact formulas to obtain the damped vibration response:

Consider the following SDOF vibratory system



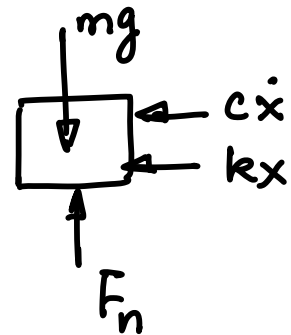
$$W = mg = 96.6 \text{ lb}$$

$$(m = 3 \text{ slugs})$$

$$k = 1 \text{ lb/in} = 12 \text{ lb/ft}$$

# SDOF damped free vibration example

FBD



Using positive convention,

$$m\ddot{x} = -c\dot{x} + kx$$

$$\Rightarrow m\ddot{x} + c\dot{x} + kx = 0 \quad (\text{free vibration})$$

The eqn of motion can be written as:

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0$$

# SDOF damped free vibration example

Define

$$\omega_n^2 = \frac{k}{m} \quad (\omega_n = \text{natural frequency})$$

$$2\mathcal{J}\omega_n = \frac{c}{m} \quad (\mathcal{J} = \text{damping ratio})$$

$$\Rightarrow \mathcal{J} = \frac{c}{2m} \sqrt{\frac{m}{k}} = \frac{c}{2\sqrt{km}}$$

$$\text{Eqn becomes:} \quad \ddot{x} + 2\mathcal{J}\omega_n \dot{x} + \omega_n^2 x = 0$$

# SDOF damped free vibration example

Solution of the response is of the form

$$x(t) = C e^{\lambda t} \quad (\text{separation of variables technique})$$

$$\Rightarrow \underbrace{(\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2)}_{\text{set equal to zero to obtain the frequency eqn.}} C e^{\lambda t} = 0$$

set equal to zero to obtain the frequency eqn.

Solving the frequency equation (recall root-finding in ME 2500) gives

$$\lambda_1, \lambda_2 = -\zeta\omega_n \pm i \underbrace{\omega_n \sqrt{1-\zeta^2}}_{\omega_d = \text{damped natural frequency}}$$

$\omega_d = \text{damped natural frequency}$

# SDOF damped free vibration example

Note that  $\mathcal{J}\omega_n = \frac{c}{2m}$

$$\omega_n \sqrt{1 - \mathcal{J}^2} = \sqrt{\frac{k}{m} - \frac{k}{m} \left( \frac{c^2}{4km} \right)} = \sqrt{\frac{k}{m} - \left( \frac{c}{2m} \right)^2}$$

Hence the general solution can be expressed as:

$$\begin{aligned} x(t) &= e^{-\mathcal{J}\omega_n t} \left[ A' e^{i(\omega_n \sqrt{1 - \mathcal{J}^2})t} + B' e^{-i(\omega_n \sqrt{1 - \mathcal{J}^2})t} \right] \\ &= e^{-\frac{c}{2m}t} \left[ A' e^{i\sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2}t} + B' e^{-i\sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2}t} \right] \end{aligned}$$

# SDOF damped free vibration example

The above form is usually difficult to use except for frequency analysis and controls formulations. An alternate solution form, which is easier to use, is:

$$x(t) = e^{-\gamma \omega_d t} \left[ A \cos \omega_d t + B \sin \omega_d t \right] \quad (1)$$

or,

$$x(t) = e^{-\frac{c}{2m} t} \left[ A \cos \left( \sqrt{\frac{k}{m} - \left( \frac{c}{2m} \right)^2} t \right) + B \sin \left( \sqrt{\frac{k}{m} - \left( \frac{c}{2m} \right)^2} t \right) \right] \quad (2)$$

# SDOF damped free vibration example

Given that at  $t=0$ ,  $x = x_0 = 6 \text{ in} = 0.5 \text{ ft}$  ;  $\dot{x} = 0$

From eqn (i)

$$0.5 = (i) [A + 0] \Rightarrow A = 0.5 \quad (\text{i.e., } A = x_0)$$

Also, differentiate (i) to get

$$\begin{aligned} \dot{x} = & -j\omega_n e^{-j\omega_n t} [A \cos \omega_d t + B \sin \omega_d t] \\ & + e^{-j\omega_n t} [-A \omega_d \sin \omega_d t + B \omega_d \cos \omega_d t] \end{aligned}$$

$$\dot{x}(t=0) = 0 \Rightarrow -j\omega_n A + B \omega_d = 0 \Rightarrow B = \frac{j\omega_n x_0}{\omega_d}$$

Complete solution is thus

$$x = x_0 e^{-j\omega_n t} \left[ \cos \omega_d t + \frac{j\omega_n}{\omega_d} \sin \omega_d t \right] \quad (3)$$

# SDOF damped free vibration example

Note that in above expression, the term

$$\frac{\zeta \omega_n}{\omega_d} = \frac{\zeta \omega_n}{\omega_n \sqrt{1-\zeta^2}} = \frac{\zeta}{\sqrt{1-\zeta^2}}$$

Hence, (3) is **not valid** for  $\zeta=1$  (critically damped case).

It can be shown that the general solution is of the form

$$x(t) = C e^{\lambda t} + D t e^{\lambda t} \quad (\lambda = -\zeta \omega_n \pm \omega_n \sqrt{1-\zeta^2} = -\omega_n)$$

Substituting the initial conditions gives

$$x(t) = x_0 \left( e^{-\omega_n t} + \omega_n t e^{-\omega_n t} \right) \quad (4)$$

# SDOF damped free vibration example

Three damping cases:

$c = 18 \text{ lb-sec/ft}, 12 \text{ lb-sec/ft}, 6 \text{ lb-sec/ft}$

Is this system under-, over-, or critically damped?

$$\textcircled{1} \quad \frac{k}{m} - \left(\frac{c}{2m}\right)^2 = \frac{12}{3} - \left(\frac{18}{2 \times 3}\right)^2 \quad [\text{check largest damping case}]$$

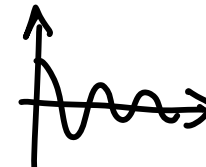
$$= 4 - (3)^2 < 0 \Rightarrow \text{over-damped}$$



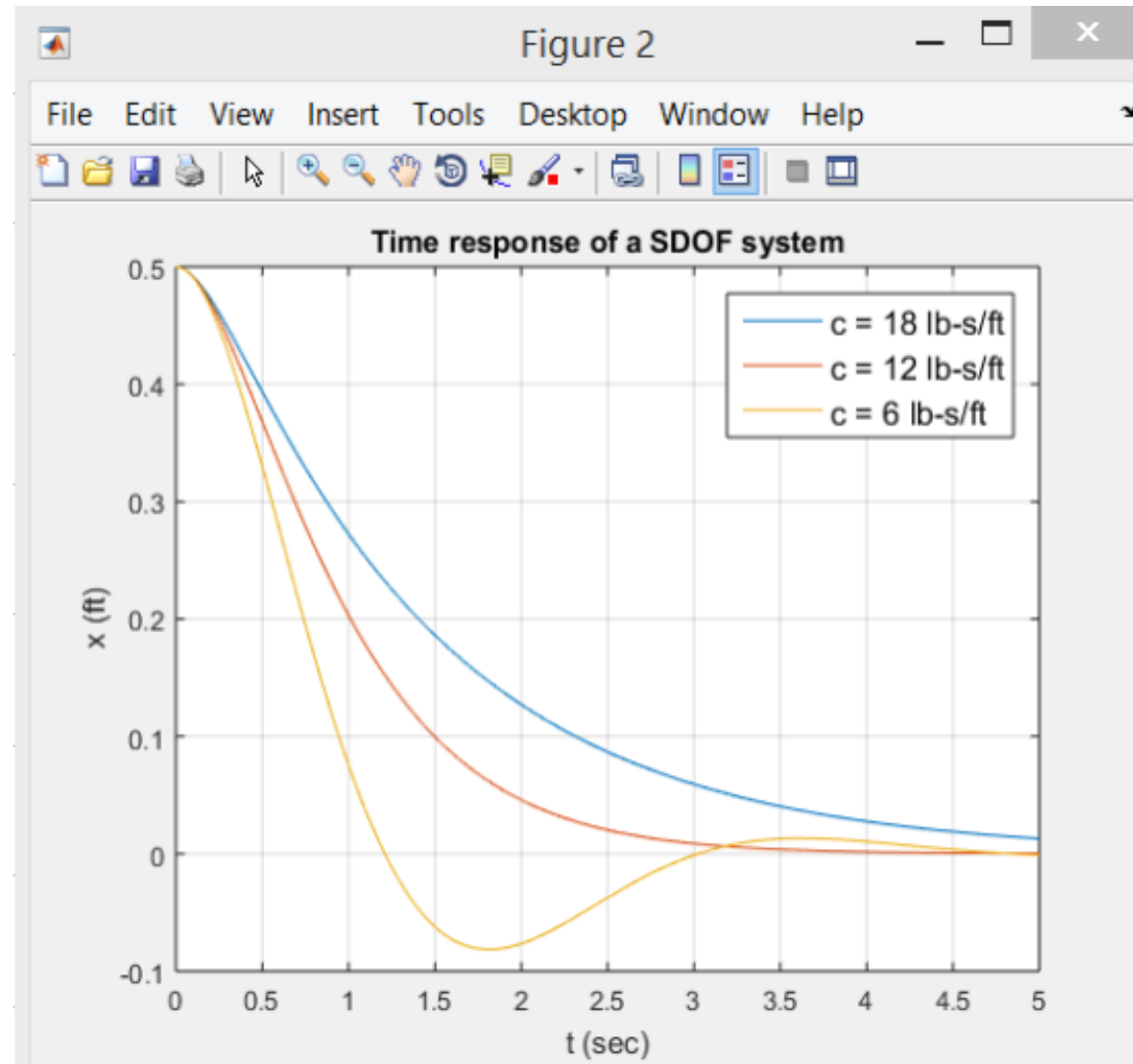
$$\textcircled{2} \quad \frac{12}{3} - \left(\frac{12}{2 \times 3}\right)^2 = 4 - (2)^2 = 0 \Rightarrow \text{critically damped}$$



$$\textcircled{3} \quad \frac{12}{3} - \left(\frac{6}{2 \times 3}\right)^2 = 4 - (1)^2 > 0 \Rightarrow \text{under-damped}$$



# SDOF damped free vibration example



## 2.1 Solution with the aid of MATLAB

- Use of MATLAB in plotting solutions.
- Solution of the governing differential equation by MATLAB ode solver.
- Build foundation for Simulink

## Inman Example 1.3.2

The human leg has a *measured* natural frequency of around 20 Hz when in its rigid (knee locked) position, in the longitudinal direction (i.e., along the length of the bone) with a damping ratio of  $\zeta = 0.224$ . Calculate the response of the tip of the leg bone to an initial velocity of  $v_0 = 0.6$  m/s and zero initial displacement (this would correspond to the vibration induced while landing on your feet, with your knees locked from a height of 18 mm) and plot the response. What is the maximum acceleration experienced by the leg assuming no damping?

**This problem is useful for design of shoes for optimal absorption of impact energy resulted from jumping.**

# Solution to Inman Example 1.3.2

$$\omega_n = \frac{20 \text{ cycles}}{1 \text{ s}} \times 2\pi \frac{\text{rad}}{\text{cycle}} = 125 \text{ rad/s}$$

$$\omega_d = 125.66 \sqrt{1 - (0.224)^2} = 122.467 \text{ rad/s}$$

$$A = \frac{\sqrt{(0.6 + (0.224)(125.66)(0))^2 + (0)(122.467)^2}}{122.467} = 0.005 \text{ m}$$

$$\phi = \tan^{-1} \left( \frac{(0)(\omega_d)}{v_0 + \zeta \omega_n(0)} \right) = 0$$

$$\Rightarrow x(t) = 0.005 e^{-28.148t} \sin(122.467t)$$

$\zeta < 1$ , thus this is an underdamped motion

Recall:

$$x(t) = e^{-\zeta \omega_n t} \left( x_0 \cos \omega_d t + \frac{v_0 + \zeta \omega_n x_0}{\omega_d} \sin \omega_d t \right)$$

## Inman Example 1.3.2 :

### Use undamped formula to get max acceleration

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_n}\right)^2}, \omega_n = 125.66, v_0 = 0.6, x_0 = 0$$

$$A = \frac{v_0}{\omega_n} m = \frac{0.6}{\omega_n} m$$

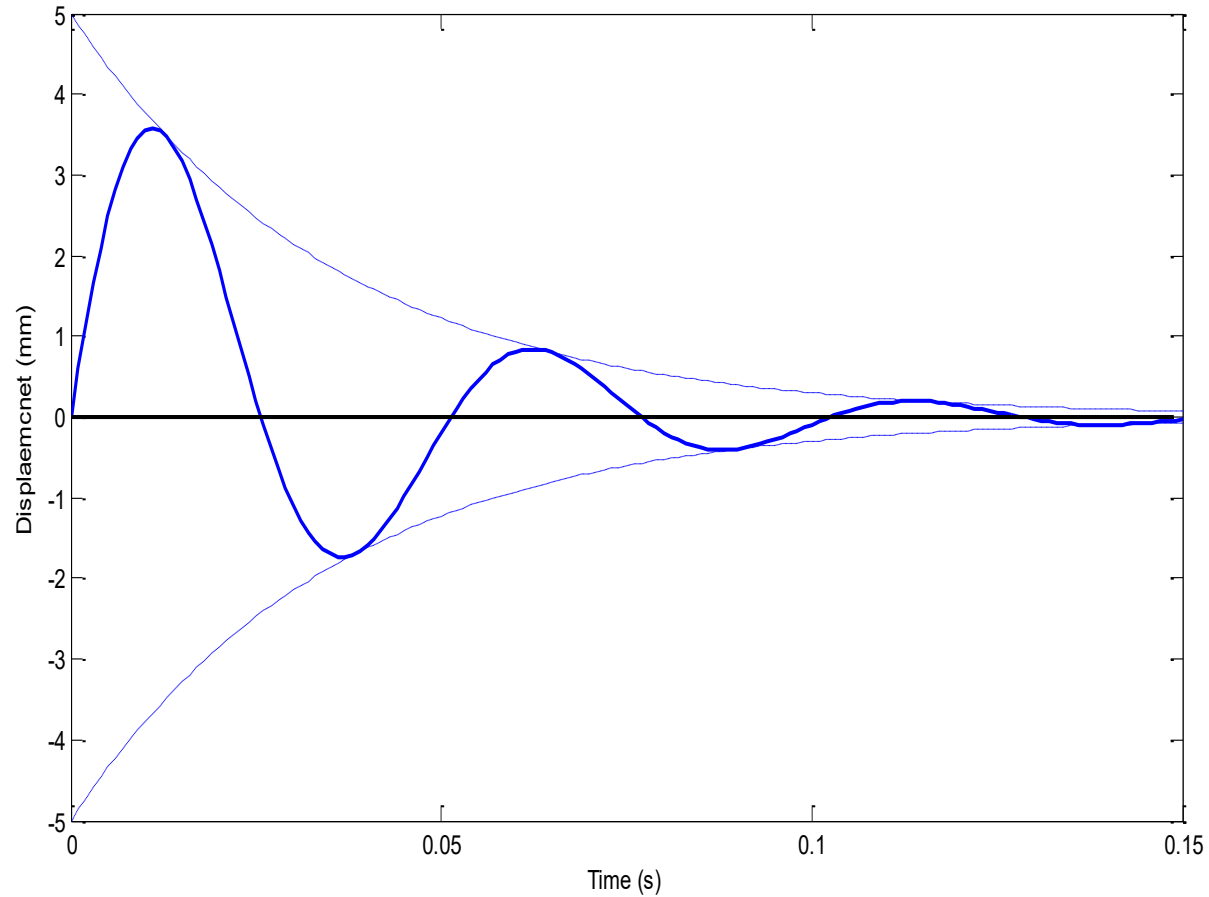
$$\max(\ddot{x}) = |-\omega_n^2 A| = \left| -\omega_n^2 \left( \frac{0.6}{\omega_n} \right) \right| = (0.6)(125.66 \text{ m/s}^2) = 75.396 \text{ m/s}^2$$

$$\text{maximum acceleration} = \frac{75.396 \text{ m/s}^2}{9.81 \text{ m/s}^2} g = 7.68 g$$

**In engineering, acceleration often expressed in terms of how many g.**

# Inman Example 1.3.2 :

## Plot of the response (practise MATLAB commands)



# Numerical solution of second-order equation of motion for vibration problems

Consider a general second-order differential equation that describes the vibration of a SDOF system given as follows

$$m\ddot{x} + c\dot{x} + kx = 0$$

We need to convert the second-order equation into two first-order equations by the [state-space formulation](#). To achieve this, define the state-space variables:  $u_1(t) = x(t)$  and  $u_2(t) = \dot{x}(t)$ . The state-space vector is  $\mathbf{u} = \{u_1 \quad u_2\}^T$

Then re-write the second-order differential equation into two first-order differential equations:

$$\begin{aligned}\dot{u}_1 &= u_2 \\ \dot{u}_2 &= -\frac{c}{m}u_2 - \frac{k}{m}u_1\end{aligned}$$

# Matrix form of the state-space equations

Combining these first-order differential equations in matrix form gives:

$$\frac{d}{dt} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}}_{\text{coefficient matrix } \mathbf{A}} \underbrace{\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix}}_{\text{state-space vector } \mathbf{u}}$$

Numerical methods such as the Euler method can then be applied to the matrix form to give:

$$\mathbf{u}(t_{i+1}) = \mathbf{u}(t_i) + \Delta t \mathbf{A} \mathbf{u}(t_i)$$

$$\mathbf{u}(t_{i+1}) = [\mathbf{I} + \Delta t \mathbf{A}] \mathbf{u}(t_i)$$

# MatLab solutions using 'ode23' and 'ode45'

- Using Runge-Kutta method. More sophisticated than the Euler method, but more accurate.
- Adaptively chooses  $\Delta t$  (i.e., if solution  $x(t)$  is rapidly changing locally, then  $\Delta t$  is chosen to be small, and visa-versa).
- Algorithm can also be used to solve nonlinear equations.

## Create MatLab function

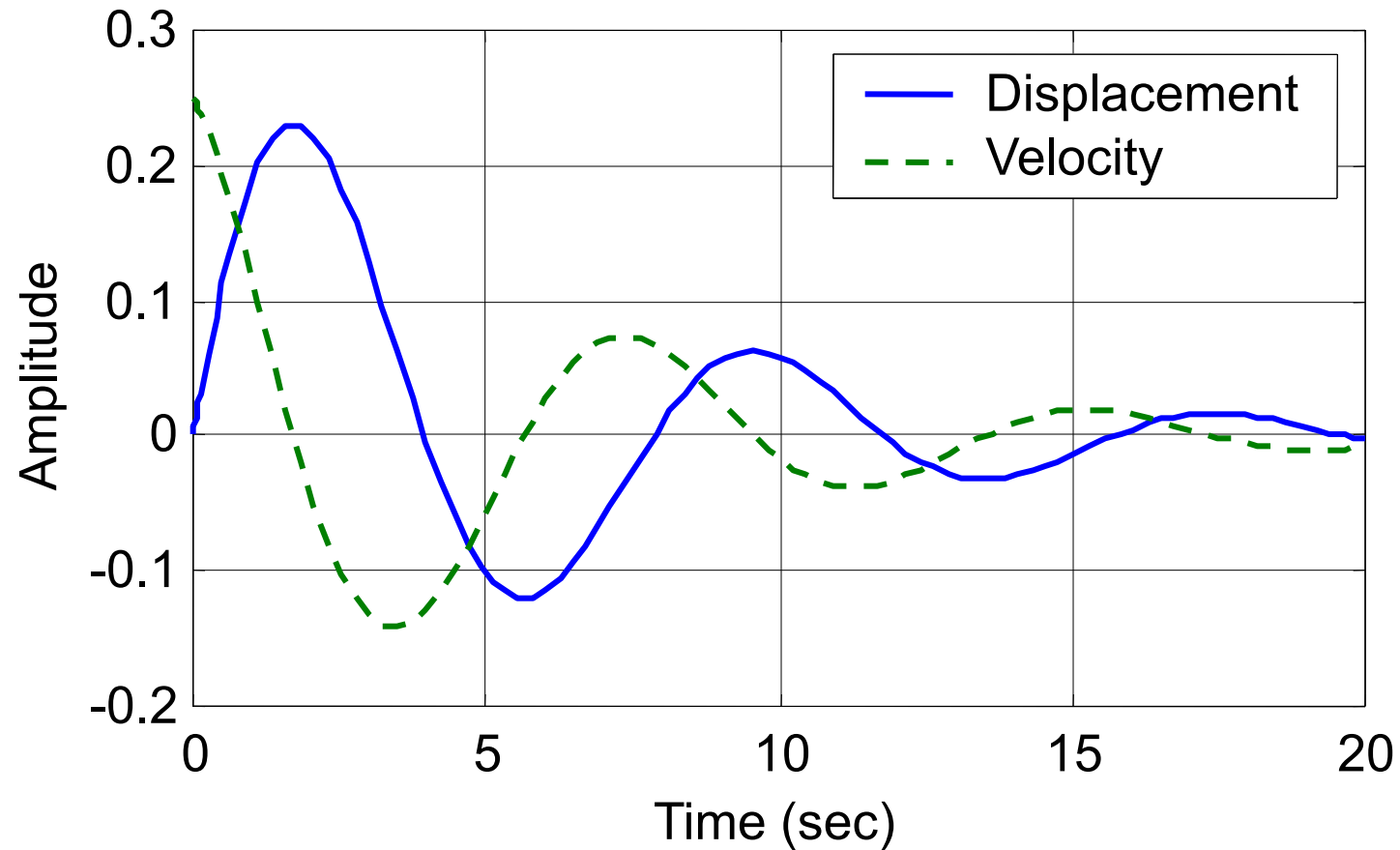
```
function udot=sdof(t,u)
k=2;c=1;m=3;
A=[0 1;-k/m -c/m];
udot=A*u;
```

Saved as sdof.m

## In the command window

```
» t0=0; tf=20;
» u0=[0, 0.25];
» [t,u]=ode45(@sdof,[t0 tf],u0);
» plot(t,u)
```

# Sample plots



## Practice problem:

### Numerical simulation of De Silva Example 2.4

Consider the simple oscillator shown in Figure 2.15, with parameters given by  $m = 4$  kg,  $k = 1.6 \times 10^3$  N/m, and two cases of damping:

1.  $b = 80$  N-s/m
2.  $b = 320$  N-s/m

Study the nature of the free response in each case.

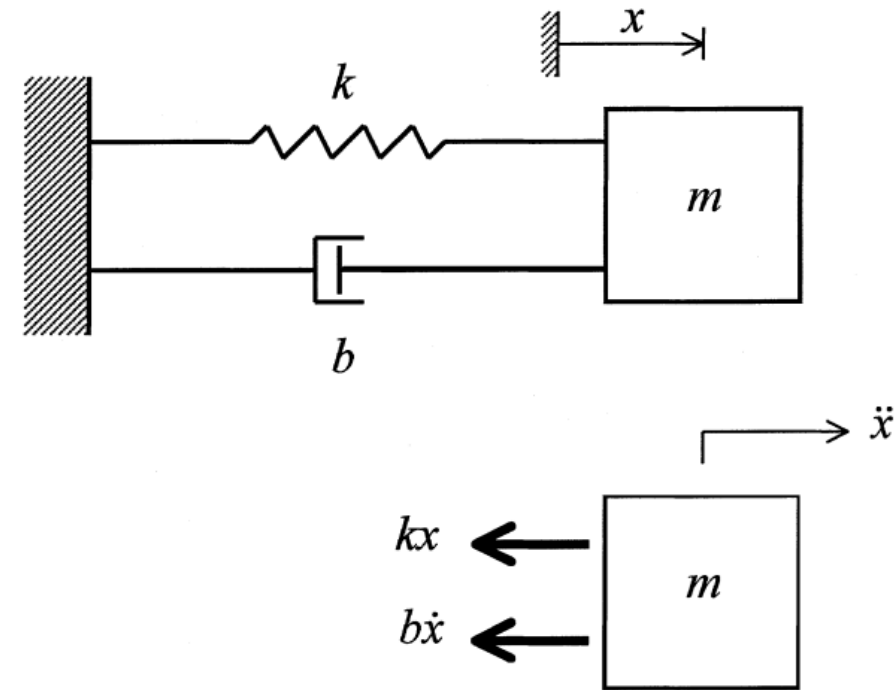


FIGURE 2.15 A damped simple oscillator and its free-body diagram.

## Practice problem: Numerical simulation of De Silva Example 2.4

Case 2: The free response is given by

$$x = C_1 e^{-5.36t} + C_2 e^{-74.64t}$$

The constants  $C_1$  and  $C_2$ , can be determined using initial conditions. Note that the second term on the RHS goes to zero much faster than the first term, as shown in Figure 2.16. Hence, the first term will dominate and will determine the dominant time constant, level of stability, and speed of response. Specifically, the response may be approximated as

$$x \cong C_1 e^{-5.36t}$$

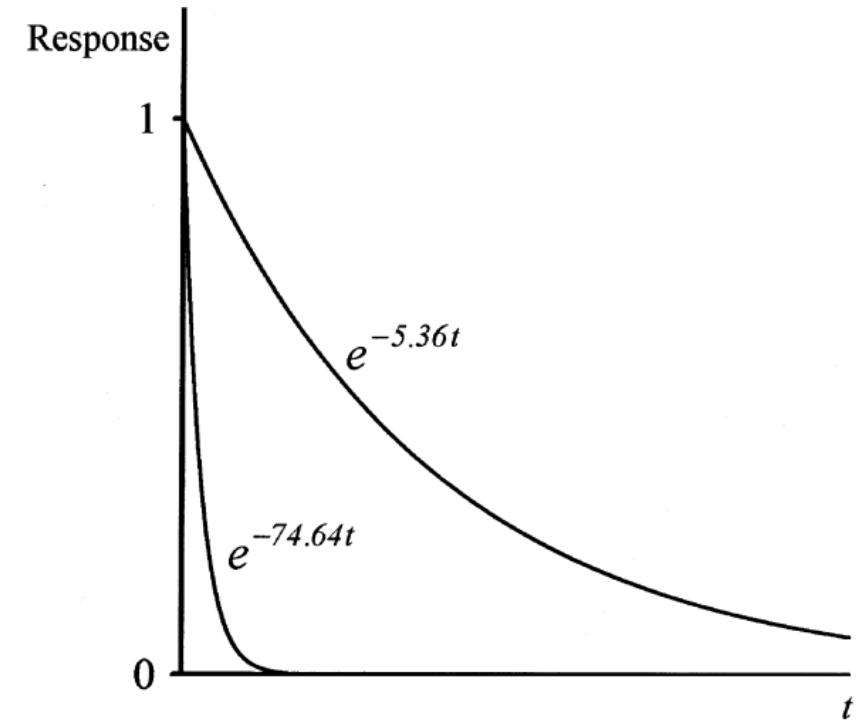


FIGURE 2.16 The free (homogeneous) response components of an overdamped system.

## Practice problem: Numerical simulation of De Silva Example 2.4

Consider the free vibration of SDOF oscillators with  $m = 4$  kg,  $k = 1.6 \times 10^3$  N/m, and damping constant:

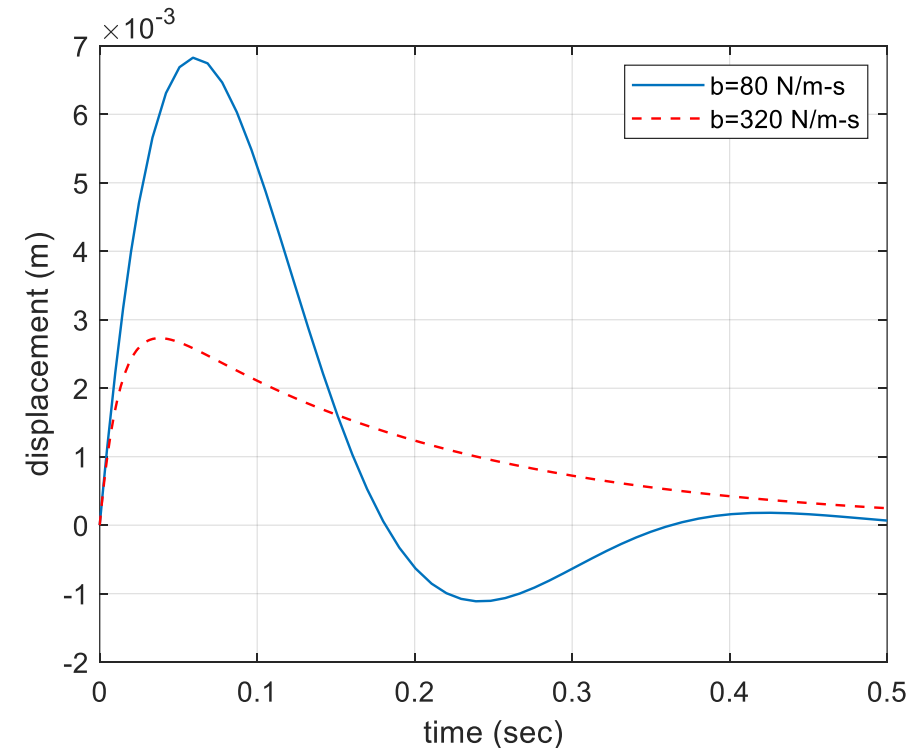
1.  $b = 80$  N–s/m
2.  $b = 320$  N–s/m

Study the nature of the response in each case through numerical simulations. Note that the damping ratio in each case is:

$$(a) \quad \zeta = \frac{b}{2\sqrt{mk}} = \frac{80}{2\sqrt{4 \times 1.6 \times 10^3}} = 0.5$$

$$(b) \quad \zeta = \frac{b}{2\sqrt{mk}} = \frac{320}{2\sqrt{4 \times 1.6 \times 10^3}} = 2$$

MATLAB code is shown as follows. It is seen that the response in case (b) is over-damped. The response for case (a) is quickly dissipated within one cycle.



## Practice problem:

### Numerical simulation of De Silva Example 2.4 using ODE solver

```
t0 = 0; tf = 0.5;
tspan = [t0 tf]; % define initial and final time for simulation
x0 = [0 0.25]; % define initial conditions
global m b k
m = 4; % mass in unit of [kg]
k = 1.6e3; % stiffness in unit of [N/m]
% Plot the response of the two cases
figure(1)
% Case 1
b = 80; % damping in unit of [N/m-s]
[t1, u1] = ode45(@DS2e_e2_5, tspan, x0);
plot(t1,u1(:,1)), hold on
% Case 2
b = 320; % damping in unit of [N/m-s]
[t2, u2] = ode45(@DS2e_e2_5, tspan, x0);
plot(t2,u2(:,1), 'r--'), hold off
xlabel('time (sec)'), ylabel('displacement (m)'), grid
legend('b=80 N/m-s', 'b=320 N/m-s')

function udot = DS2e_e2_5(t,u)
% Simulate Example 2.5 of De Silva textbook 2e (how the function is named)
% Defines the equation of motion of a SDOF oscillator
global m b k
udot = [u(2); (-k*u(1)-b*u(2))/m];
end
```

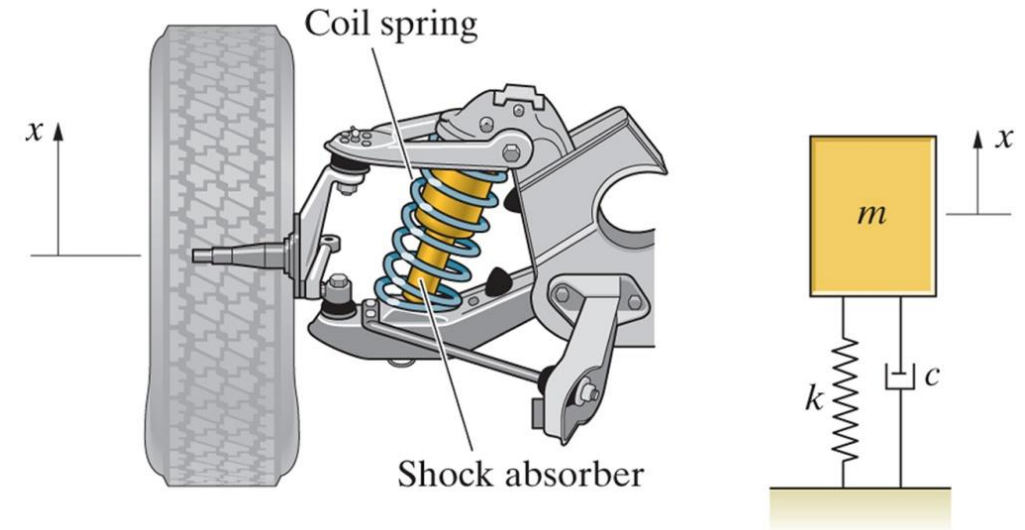
# Practice problem: Numerical simulation of Suspension

Consider the suspension modeled as a simple damped oscillator. The governing equation of motion is:

$$m\ddot{x} + c\dot{x} + kx = 0$$

Assume motion starts from rest,  $v_0 = 0$ . Unless otherwise stated, the nominal values of the mass and the spring stiffness used in numerical simulations are:  $m = 1,000$  kg,  $k = 81,000$  N/m.

Consider a disturbance  $x_0$  applied to the system initially at rest. Plot the normalized response (normalized to  $x_0$ ) of the system for three values of the damping coefficient,  $c = 3,000, 18,000$ , and  $25,000$  N · s/m on a single graph. Properly label your graph with axis labels, title, legend, etc. Obtain a similar plot for the mechanical energy (normalized to the initial energy).



```

%% Definition of system parameters:
global m c k
global x0 v0
m = 1e3; % mass in [kg] (nominal value)
k = 81e3; % stiffness in [N/m] (nominal value)
% c      % damping coefficient in [N-s/m] (different values)
x0 = 1;  % initial displacement in [m], normalized and arbitrarily set to be equal to 1.
% Note: this doesn't mean that the initial displacement is 1 m,
% which obviously doesn't exist in any engineering application.
v0 = 0;  % initial velocity in [m/s]

% Plot the time response and total mechanical energy of the SDOF system
% for the different cases of damping being considered.
% Note: Project asks for only the cases of c = [3 18 25]*1e3
% c = 3e3 N.s/m % underdamped case (zeta = 0.167)
% c = 18e3 N.s/m % critically damped case (zeta = 1)
% c = 25e3 N.s/m % overdamped case (zeta = 1.389)
c_values = [3 18 25]*1e3;
t_data=linspace(0,2,101);
% Plot of the time response
figure
for j = 1:length(c_values)
    c = c_values(j);
    % plot the free response normalized with respect to x0
    x_data=frsp_s dof(t_data)/x0;
    plot(t_data,x_data),hold on
end
hold off

```

# Practice problem: Numerical simulation of Suspension

```
% Set plot parameters
ax = gca;
ax.YLim = [-0.8,1];
ax.XTick = (0:0.2:2);
ax.YTick = (-0.8:0.2:1);
ax.FontSize = 11;
ax.LineWidth = 1;
grid, xlabel('t [sec]'), ylabel('x/x_0 (normalized response)')
title({'Time history of the free response of a suspension system', ...
'modeled by a single-degree-of-freedom system'})
LEG = legend('c = 3e3 N-s/m', 'c = 18e3 N-s/m', 'c = 25e3 N-s/m');
set(LEG, 'FontSize', 12)
```

# Practice problem: Numerical simulation of Suspension

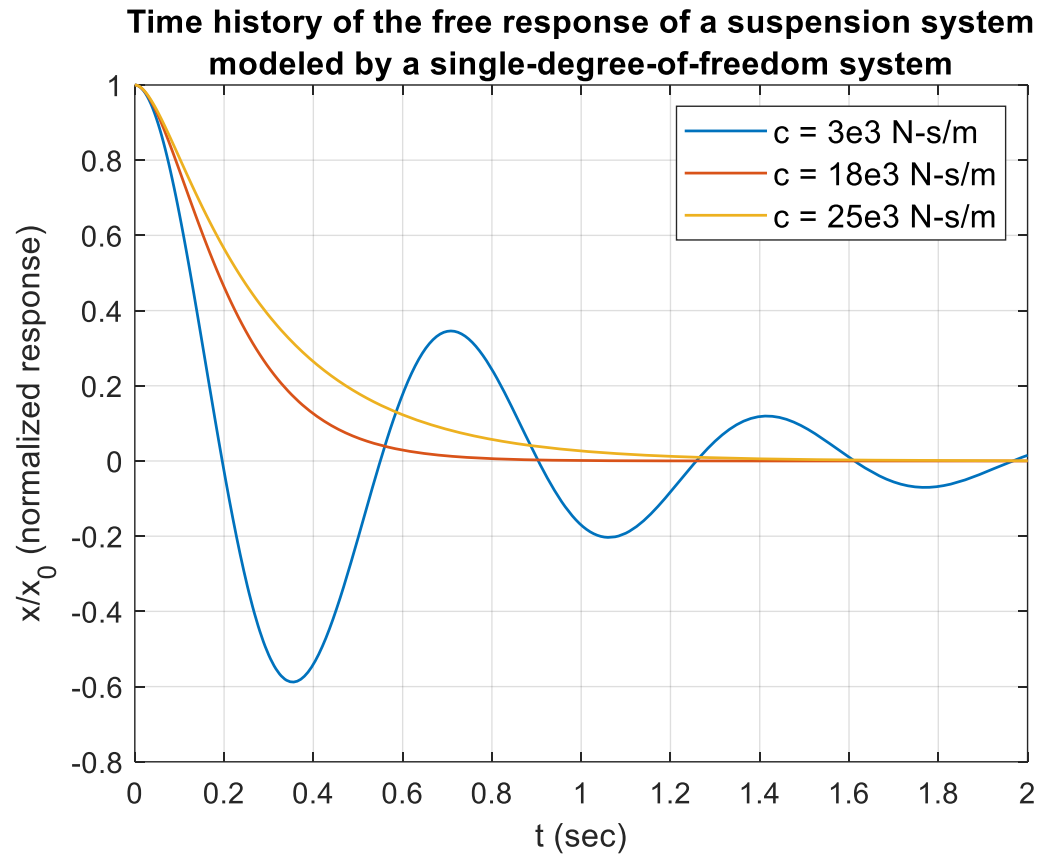
```
function x = frsp_s dof(t)
% Free response function of a SDOF system
global m c k
global x0 v0
% m = mass in [kg or slug]
% c = damping coefficient in [N-s/m or lb-s/ft]
% k = stiffness in [N/m or lb/ft]
% x0 = initial displacement in [m or ft]
% v0 = initial velocity in [m/s or ft/s]

wn = sqrt(k/m); % natural frequency ("omega_n")
zeta = c/2/sqrt(k*m); % damping ratio
wd = wn*sqrt(1-zeta^2); % damped natural frequency ("omega_d")

% Students need to complete the rest of this function code
```

# Response under different damping ratios

- Sample MATLAB simulation. Choose  $m = 1,000$  kg,  $k = 81,000$  N/m,  $v_0 = 0$ .



Motions exhibit energy dissipation:

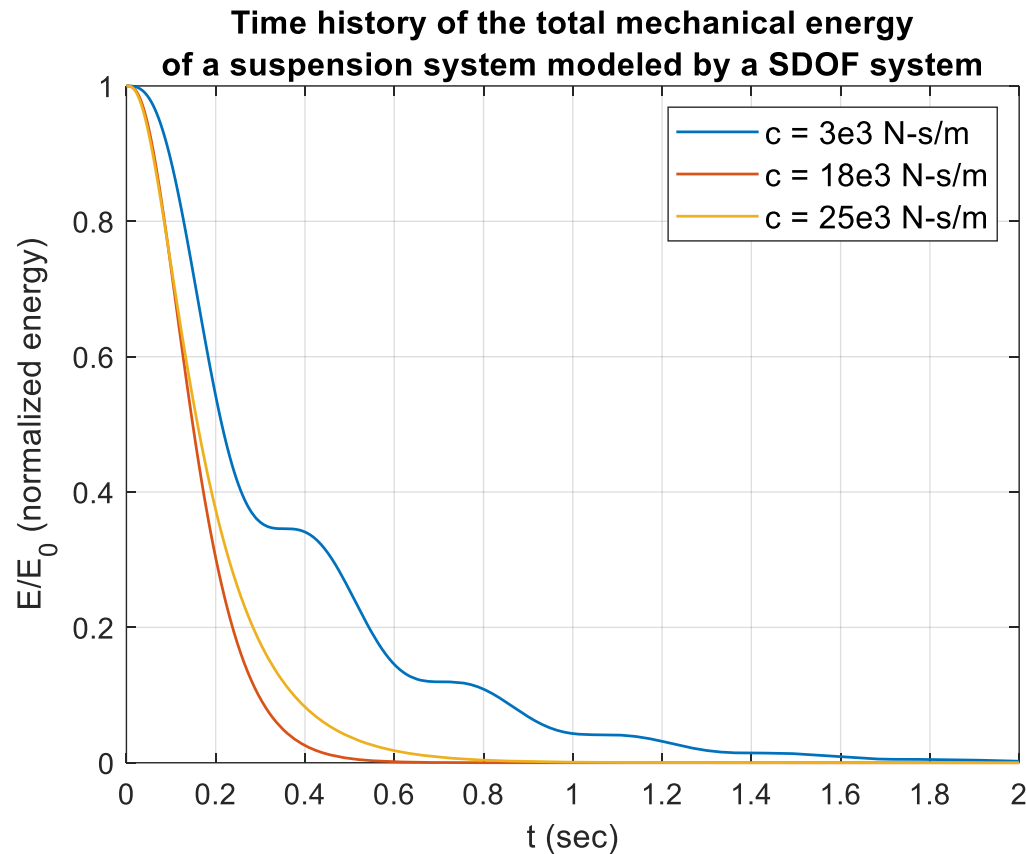
$\zeta < 1$ , oscillatory

$\zeta \geq 1$ , purely decaying

$c = 18,000$  N-s/m is the critically damped case for this example.

# Energy dissipation under different damping ratios

- Sample MATLAB simulation. Choose  $m = 1,000$  kg,  $k = 81,000$  N/m,  $v_0 = 0$ .



Energy dissipates uniformly over time. Critically damped case seems to dissipate energy most optimally. For optimisation problem, examine an energy functional called action:

$$action = \int_0^{t^*} E(t) dt$$

where,  $t^*$  is a user-defined design parameter.

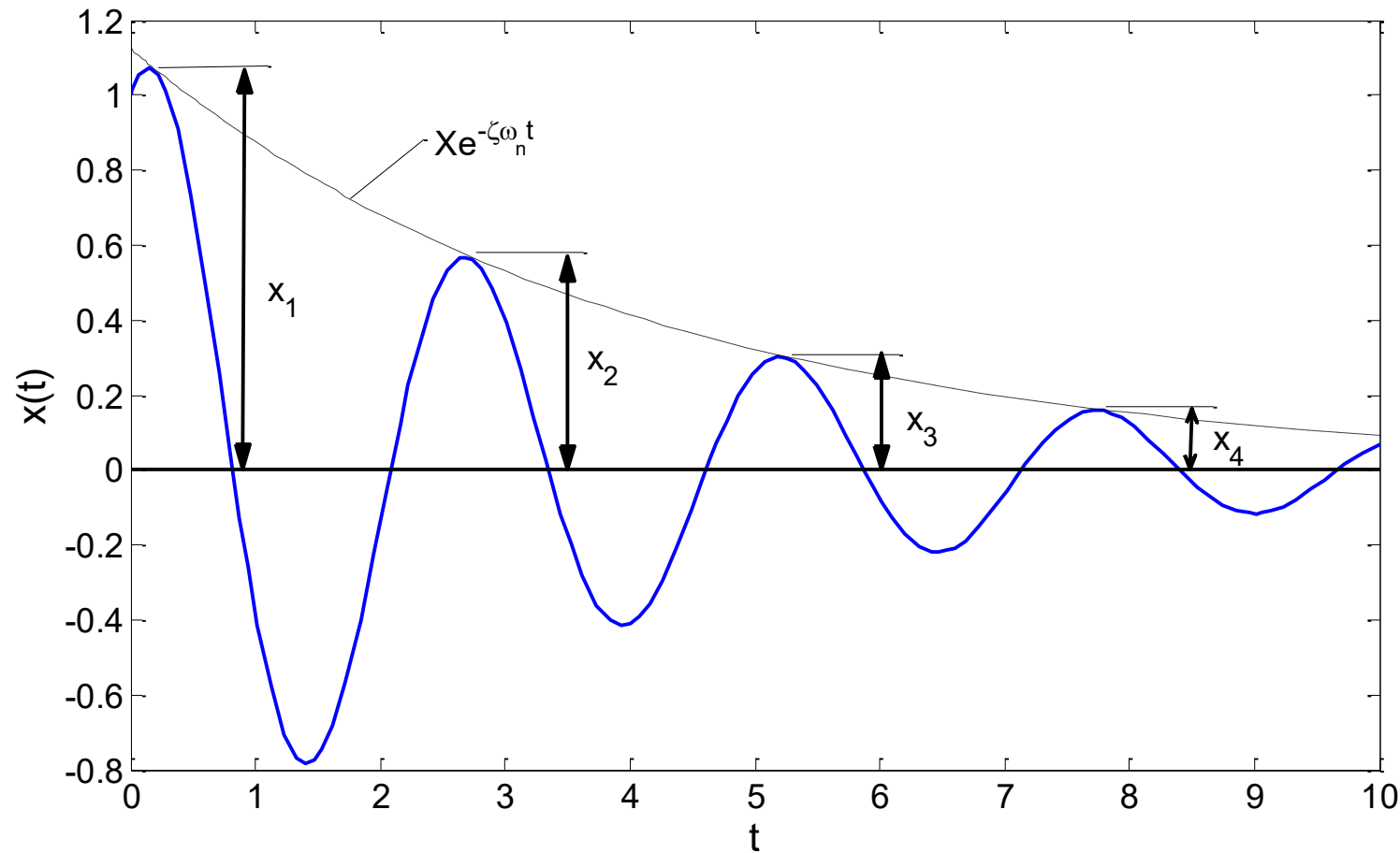
$c = 18,000$  N-s/m is the critically damped case for this example.

## 2.2 Damping Measurements under free response

- Unlike the measurement of the spring constant by static tests (tensile tests), measurement of the damping constant requires dynamic tests.
- The logarithmic decrement method is a time domain method based on the free vibration.
- There is another method called the half-power method based on the forced response under harmonic excitation. This will be presented in later content.

# Logarithmic Decrement Method

Logarithmic decrement method for determining the damping ratio of underdamped systems



# Logarithmic Decrement Method

```
% Script file: log_decay
% Sample plot to demonstrate the idea of logarithm decay
% Plot and data in folder: ../ME5410/Lecture Notes
% Simulate the following equation as an illustration for logarithmic decrement
figure
%
omega_n = 2.5; zeta = 0.1;
% Initial conditions: x(0) = 1, xdot(1) = 1;
[t,y] = ode23(@(t,u) [u(2); -2*zeta*omega_n*u(2)-omega_n^2*u(1)],...
    [0,10],[1,1]);
plot(t,y(:,1)) % plot the response solution obtained using ode23
hold on
plot([0,10],[0,0],'k') % generate the zero horizontal line on plot
hold on
x0 = 1; v0 = 1; % initial conditions
omega_d = omega_n*sqrt(1-zeta^2);
X = sqrt(((v0+omega_n*zeta*x0)^2+(x0*omega_n)^2)/omega_d^2);
% plot the decayed response envelope curve
fplot(@(t) X*exp(-zeta*omega_n*t),[0,10], '--k')
```

# Logarithmic Decrement Method

The damping ratio  $\zeta$  can be experimentally determined from the free response by the logarithmic decrement method. To illustrate this approach, recall the period of damped oscillations is  $T_d = 2\pi/\omega_d$ . Thus, we have

$$\frac{x(t)}{x(t + nT)} = \frac{Ae^{-\zeta\omega_n t} \sin(\omega_d t + \phi)}{Ae^{-\zeta\omega_n(t+nT)} \sin[\omega_d(t + nT) + \phi]}$$

Since,

$$\sin[\omega_d(t + nT) + \phi] = \sin(\omega_d t + \phi + 2n\pi) = \sin(\omega_d t + \phi)$$

Hence,

$$\frac{x(t)}{x(t+nT)} = \frac{e^{-\zeta\omega_n t}}{e^{-\zeta\omega_n(t+nT)}} = e^{\zeta\omega_n nT} \quad (1)$$

# Logarithmic Decrement Method

Define the logarithmic decrement. From Eq. (1):  $\zeta \omega_n nT = \ln \left[ \frac{x(t)}{x(t+nT)} \right]$ . Since  $\omega_n T = \omega_n \frac{2\pi}{\omega_d} = \frac{\omega_n 2\pi}{\sqrt{1-\zeta^2} \omega_n} = \frac{2\pi}{\sqrt{1-\zeta^2}}$ . Let  $r = x(t)/x(t+nT)$ , the decrement, one then obtains a relation for the logarithmic decrement  $\ln r$

$$\frac{2\pi n \zeta}{\sqrt{1-\zeta^2}} = \ln r \Rightarrow \frac{\zeta}{\sqrt{1-\zeta^2}} = \frac{1}{2\pi n} \ln r = \alpha \quad (2)$$

Note that  $\frac{1}{n} \ln r$  is the "per-cycle" logarithmic decrement, and  $\frac{\ln r}{2\pi n}$  is the "per-radian" logarithmic decrement. From Eq. (2), one has

$$\zeta = \sqrt{\alpha^2 / (1 + \alpha^2)}$$

This is the basis of the logarithmic decrement method of measuring damping. Start by measuring a point  $x(t)$  and another point  $x(t+nT)$  at  $n$  cycles later. For high accuracy, pick the peak points of the response curve for the measurement of  $x(t)$  and  $x(t+nT)$ . From Eq. (2), for **small damping**,  $\zeta \approx \alpha$ , which is the per-radian logarithmic decrement.